Introduction

1.4 Measures of Fluid Mass and Weight

Density
Density is defined as the mass per unit volume of a substance.

- BG system: $[\rho] = \frac{\text{slug}}{\text{ft}^3}$
- EE system: $[\rho] = \frac{\text{lbm}}{\text{ft}^3}$
- SI system: $[\rho] = \frac{\text{kg}}{\text{m}^3}$

Density of fluids are dependent on both temperature and pressure but for liquids, variations in temperature and pressure have a negligible effect on density. For gases, however, density is highly dependent on both temperature and pressure.

Specific Volume
Defined as volume per unit mass of a substance, and is the reciprocal of density:

$$\cancel{v} = \frac{1}{\rho} = \frac{\cancel{V}}{m}$$

More common in thermodynamics (see spring 2026 with KAMAU WRIGHT) than fluid mechanics.

Specific Weight
Defined as weight per unit volume of a substance:

$$\gamma = \rho g$$

Where g is the *local* acceleration of gravity:

- $g = 32.174\ \frac{\text{ft}}{\text{s}^2}$ (BG)
- $g = 9.807\ \frac{\text{m}}{\text{s}^2}$ (SI)

It has units of

- BG system: $[\gamma] = \frac{\text{lbf}}{\text{ft}^3}$
- SI system: $[\gamma] = \frac{\text{N}}{\text{m}^3}$

Specific Gravity
Defined as the ratio of fluid density to water at 4 °C or 39.2 °F:

$$\text{SG} = \frac{\rho}{\rho_{H_2O\,@4^\circ\text{C}}}$$

The density of water at room temperature, $\rho_{H_2O\,@4^\circ C}$ is:

- BG system: $\rho_{H_2O\,@4^\circ C} = 1.94\,\frac{\text{slug}}{\text{ft}^3}$
- SI system: $\rho_{H_2O\,@4^\circ C} = 1000\,\frac{\text{kg}}{\text{m}^3}$

Conversions:
- $1\,\text{slug} = 32.174\,\text{lbm}$
- $1\,\text{lbf} = 1\,\frac{\text{slug ft}}{\text{s}^2}$
- $1\,\text{slug} = 14.5939\,\text{kg}$
- $1\,\text{m} = 3.28084\,\text{ft}$

1.5 Ideal Gas Law (Equation of State)

$$p\cancel{v} = RT$$

Where:

- p: absolute pressure of the gas
- $\cancel{v}$: specific volume of the gas
- R: Gas constant for the gas
- T: absolute temperature of the gas

Alternative form:

$$p = \rho RT$$

Constant Volume Process
For example, processes within a rigid container ($\cancel{\forall}$, R, and m are constant):

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Pressure

Pressure is defined as force per unit area and has units of

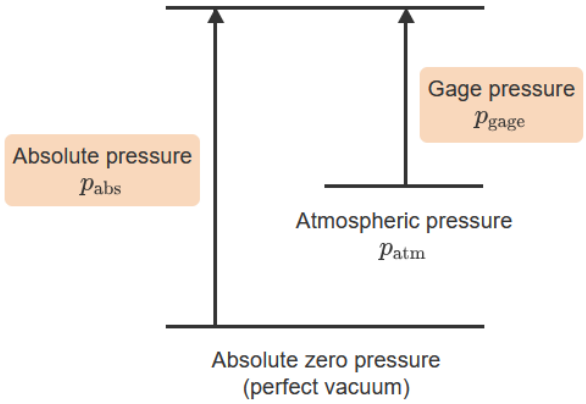
- BG system: $[\text{pressure}] = \frac{\text{lbf}}{\text{ft}^2}$ (psf)
- EE system: $[\text{pressure}] = \frac{\text{lbf}}{\text{in}^2}$ (psi)
- SI system: $[\text{pressure}] = \text{Pa} = \frac{\text{N}}{\text{m}^2}$ (Pa)

Absolute vs Gage Pressure
Measured relative to absolute zero pressure (perfect vacuum). While gage pressure is measured relative to atmospheric pressure. Thus, the relationship between absolute and gage pressure is:

$$(\text{Gage Pressure}) + (\text{Atmospheric Pressure}) = (\text{Absolute Pressure})$$

Where atmospheric pressure is:

- EE system: $p_{atm} = 14.7\,\text{psi}$
- SI system: $p_{atm} = 101\,\text{kPa}$



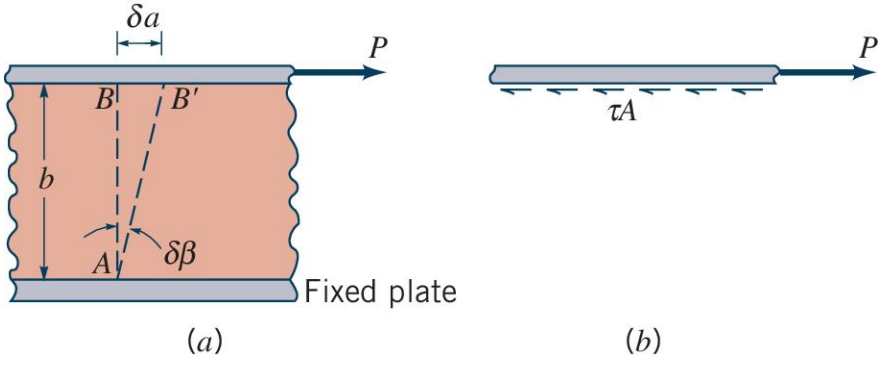
Conversions:

- $\text{K} = {}^\circ\text{C} + 273.15$
- $\text{R} = {}^\circ\text{F} + 460$

1.6 Viscosity

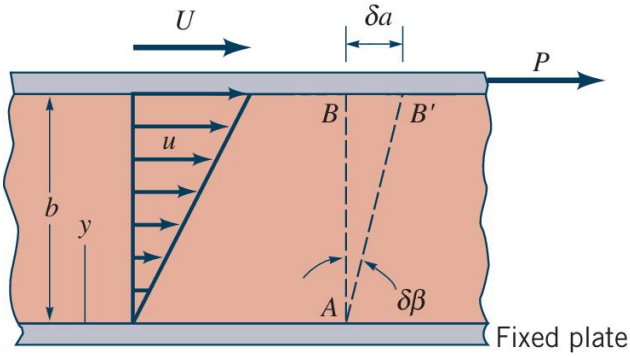
Viscosity describes the "fluidity" of a fluid, or how easily a fluid flows. It is a measure of the internal resistance of a fluid to flow. The higher the viscosity, the more resistant the fluid is to flow.

Consider the following set up with two parallel plates where the bottom plate is fixed, and the top plate is pulled by a load P. In between the plates is a solid (ie. steel) fixed to both plates.



Under the load P, the steel will deform, and the top plate will be displaced a small δa. AB will be rotated by a small angle $\delta\beta$ to AB'. We call $\delta\beta$ the shearing strain.

Now consider if a *fluid* is placed in between the plates instead of a solid. When a load P is applied, the top will now move continuously with a velocity U instead of stopping after a displacement δa. This is due to the property of fluids that they cannot resist a shearing stress, and thus they will deform continuously under a shearing stress.



The fluid in contact with the upper plate has the same velocity U and the fluid in contact with the lower plate has zero velocity. Thus, in between the plates, the fluid has a linear velocity profile:

$$u(y) = \frac{U}{b}y$$

The fluid effectively "sticks" to the solid boundaries, this is known as the no-slip condition.

The velocity gradient is then defined by:

$$\frac{du}{dy}$$

Rate of Shearing Strain

In a small time increment δt, the top plate moves a distance δa and the line AB rotates by a small angle $\delta\beta$. The shearing strain is then defined as:

$$\delta\beta \approx \tan\delta\beta = \frac{\delta a}{b} \quad \text{(by the small angle approx)}$$

Since $\delta a = U\delta t$, we can express the shearing strain as:

$$\delta\beta = \frac{U\delta t}{b}$$

Notice that $\delta\beta$ is a function of the velocity U (which is dependent on P) and time.

The rate of shearing strain $\dot{\gamma}$ is then defined as the limit as $\delta t \to 0$:

$$\dot{\gamma} = \lim_{\delta t \to 0} \frac{\delta\beta}{\delta t}$$

For the linear velocity profile given in the example above, the rate of shearing strain is $\dot{\gamma} = \frac{U}{b} = \frac{du}{dy}$.

Viscosity (omg that's the section name)

Further experimentation would find that the shearing stress τ is proportional to the rate of shearing strain $\dot{\gamma}$. Thus, for common fluids (such as water, oil, gasoline, and air), we can state the relationship:

$$\tau = \mu\dot{\gamma}$$

Where μ is the dynamic/absolute viscosity of the fluid.

Units of Viscosity

The units of dynamic viscosity are FTL^{-2}:

- BG & EE system: $[\mu] = \frac{\mathrm{lbf\,s}}{\mathrm{ft}^2}$
- SI system: $[\mu] = \mathrm{Pa\,s} = \frac{\mathrm{N\,s}}{\mathrm{m}^2}$

Types of Fluid

Different types of fluids can be classified based on the relationship between shearing stress and rate of shearing strain:

- **Newtonian Fluids** have a linear relationship between shearing stress and rate of shearing strain du/dy. (Ex: water, air, gasoline, and oil)
- **Shear-Thinning Fluids** have a non-linear relation with the rate of shearing strain, where the apparent viscosity decreases as shear rate increases. In other words, μ is not constant. (Ex: latex paint, not yummy...)
- **Shear-Thickening Fluids** have a non-linear relation with the rate of shearing strain, where the apparent viscosity increases as shear rate increases. (Ex: cornstarch in water, crunchy)
- **Bingham Plastics** behave like a solid at low shear stress, but flow like a fluid at high shear stress. It doesn't flow until yield stress is exceeded. (Ex: toothpaste, ketchup, and mayonnaise)

Effect of Temperature on Viscosity

The effect of temperature on viscosity can be approximated by the following empirical relationship called the *Sutherland Equation* for gases:

$$\mu = \frac{CT^{3/2}}{T + S}$$

Where C and S are constants for a given gas and T is the absolute temperature.

For liquids, the empirical relationship that is used is the *Andrade Equation*:

$$\mu = De^{B/T}$$

Where D and B are constants for a given liquid and T is the absolute temperature.

Kinematic Viscosity

The kinematic viscosity ν is defined as the ratio of dynamic viscosity to density. Often dynamic viscosity appears in fluid flow problems together with density, thus it is convenient to define a placeholder variable for this ratio:

$$\nu = \frac{\mu}{\rho}$$

It has units of L^2/T

Reynold's Number

This is a dimensionless number that describes the ratio of inertial forces to viscous forces in a fluid flow:

$$\mathrm{Re} = \frac{\rho V D}{\mu}$$

Where

- ρ: density of the fluid
- V: mean velocity of the fluid
- D: characteristic length (such as diameter of a pipe)
- μ: dynamic viscosity of the fluid

1.7 Compressibility of Fluids

Bulk Modulus E_ν

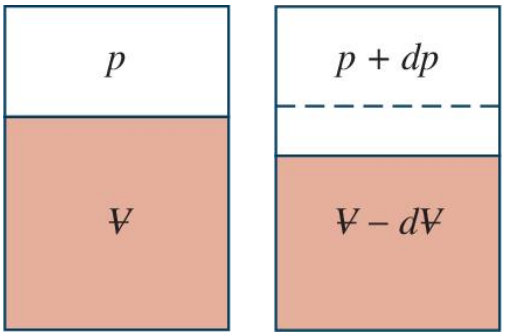
Describes how easily the volume of a fluid can be compressed. Keeping mass constant, this also describes how easily the density of a fluid can be changed under a change in pressure applied. It is defined as

$$E_\nu = -\frac{dp}{d\cancel{V}/\cancel{V}}$$

Or alternatively, in terms of density. Since $\cancel{V} = m/\rho$, a decrease in volume corresponds to an increase in density:

$$E_\nu = \frac{dp}{d\rho/\rho}$$

Where dp is the change in pressure to cause a $d\cancel{V}$ change in volume, or a $d\rho$ change in density. The first equation is negative since an increase in pressure results in a decrease in volume.



Units
It has units of pressure FL^{-2}

For a large bulk modulus, the fluid is incompressible, and for a small bulk modulus, the fluid is compressible.

Gas Compression and Expansion Processes

Constant Temperature Process (Isothermal)
Derived from the ideal gas law:

$$\frac{p}{\rho} = \text{constant}$$

Constant Entropy Process (Isentropic)
For a frictionless process with no heat exchange with surroundings:

$$\frac{p}{\rho^k} = \text{constant}$$

Where k is the ratio of specific heats $k = c_p/c_{\forall}$. c_p is the specific heat at constant pressure, and $c_{\forall}$ is the specific heat at constant volume. Additionally, the specific heats are related by the gas constant R:

$$R = c_p - c_{\forall}$$

Rewriting Bulk Modulus Relations
Given the above relations between density and pressure for isothermal and isentropic processes, we can rewrite the bulk modulus as:

$$E_\nu = p \quad \text{(Isothermal)}$$

$$E_\nu = kp \quad \text{(Isentropic)}$$

Note: Since the pressure and density relations were derived from the ideal gas law, these relations are only valid for ideal gases and pressures are always absolute pressures.

Speed of Sound c

As a result of the compressibility in fluids, localized pressure disturbances require non-zero time to propagate through the fluid and travels at a finite speed. This is known as the speed

of sound c and is related to the bulk modulus and density of the fluid:

$$c = \sqrt{\frac{dp}{d\rho}} = \sqrt{\frac{E_\nu}{\rho}}$$

Since the disturbance is a small change in pressure and density, there is negligible heat transfer so it can be approximated as an isentropic process where $p/\rho^k = \text{constant}$. Thus, the speed of sound can be expressed as:

$$c = \sqrt{\frac{kp}{\rho}} = \sqrt{kRT} \quad \text{(Ideal Gas Law)}$$

Additionally, the mach number relates the speed of sound to an object moving through the fluid at velocity V:

$$\text{Ma} = \frac{V}{c}$$

1.9 Surface Tension

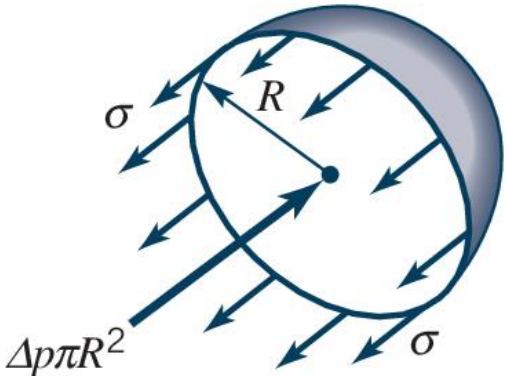
Surface tension exists at a liquid-gas interface.

Surface tension σ has units of force per unit length (FL^{-1}):

- BG system $[\sigma] = \frac{\text{lbf}}{\text{ft}}$
- SI system $[\sigma] = \frac{\text{N}}{\text{m}}$

Surface Tension and Pressure Within a Drop of Fluid

Consider the following free-body diagram of a cross-section of a spherical drop of fluid. The surface tension acts along the edge to shrink the drop's surface area while the pressure within the drop acts to expand the drop's surface area. At equilibrium, the two forces are equal:



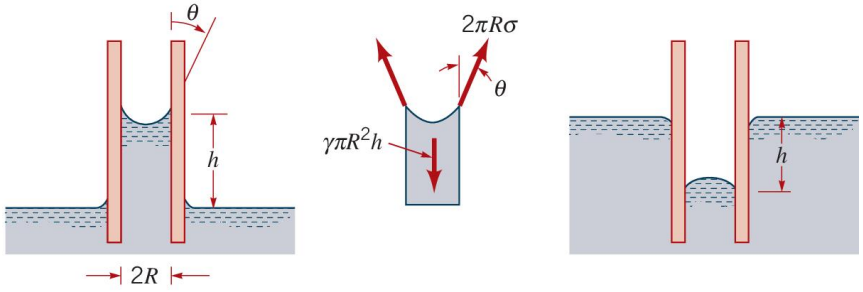
Force due to surface tension: $2\pi R\sigma$
Force due to pressure within the drop: $\Delta p \pi R^2$ where Δp is the pressure difference across the drop's surface ($p_{\text{inside}} - p_{\text{outside}}$)

Setting the two forces equal:

$$2\pi R\sigma = \Delta p \pi R^2 \implies \Delta p = p_i - p_o = \frac{2\sigma}{R}$$

Capillary Action in smoll Tubes

In the figure below, (a) shows a rise in liquid level for a liquid that wets the tub (hydrophilic) and (c) shows a drop in liquid level for a nonwetting liquid (hydrophobic). The rise or drop in liquid level is due to the surface tension acting along the circumference of the tube. The free body diagram of the liquid for case (a) is shown in (b).



Since surface tension exists along the circumference, its magnitude is given by: $2\pi R\sigma$. With a vertical component of $2\pi R\sigma \cos\theta$ Where θ is the contact angle between the liquid and the tube.

The weight of the liquid column is given by: $\gamma\pi R^2 h$ where h is the height of the liquid column.

Setting the two forces equal at equilibrium to determine the height of the liquid column as a function of surface tension and tube geometry:

$$\sum F_y = 0 = 2\pi R\sigma\cos\theta - \gamma\pi R^2 h$$

$$\implies h = \frac{2\sigma\cos\theta}{\gamma R}$$

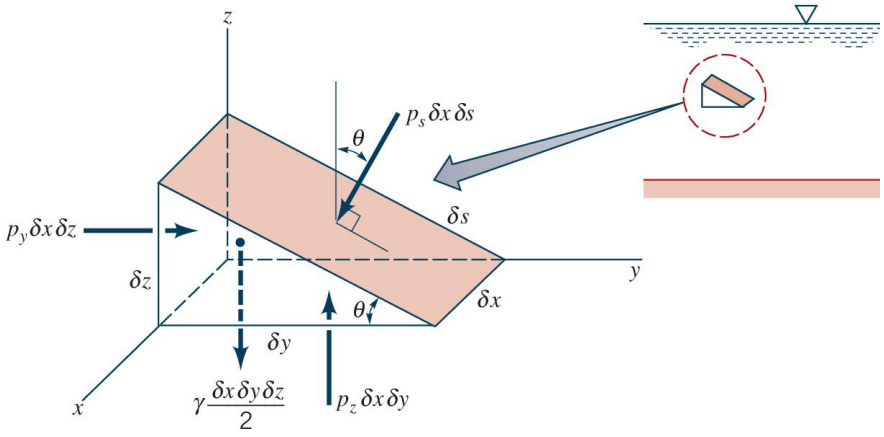
Note that θ is dependent on both the liquid and tube material properties. $\theta \approx 0^\circ$ for water in clean glass.

Fluid Statics

2.1 Pressure at a Point

Pressure is defined as the normal force per unit area at a given point acting on a given plane. However, does the orientation of the plane matter? In other words, at the same point, does pressure depend on the orientation/direction of the plane?

Consider the following small triangular wedge of fluid mass. *Since there are no shearing stresses, the only external forces acting on the mass is pressure and gravity.* (Note that the forces in the x direction is not shown for simplicity)



Note that:

- The forces due to pressure are defined as the average pressure in each direction multiplied by the area of the surface in that direction. (Ex: $F_x = p_y A_y = p_y \delta x \delta z$)
- The weight of the fluid mass is given by $W = \gamma \cancel{V} = \gamma \left(\frac{\delta x \delta y \delta z}{2}\right)$. Where volume is the volume of the triangular wedge.
- The resultant force on the cut plane is given by $F_s = p_s A_s = p_s \delta s \delta x$. Where δs is the length of the cut plane.

By Newton's second law in the y and z directions, we can write the following equations:

$$\sum F_y = p_y \delta z \delta z - p_s \delta x \delta s \sin\theta = \rho \left(\frac{\delta x \delta y \delta z}{2}\right) a_y$$

$$\sum F_z = p_z \delta x \delta y - p_s \delta x \delta s \cos\theta - \gamma \left(\frac{\delta x \delta y \delta z}{2}\right) = \rho \left(\frac{\delta x \delta y \delta z}{2}\right) a_z$$

It follows from geometry that

$$\delta y = \delta s \cos\theta, \quad \delta z = \delta s \sin\theta$$

So the equations of motion is rewritten as:

$$p_y - p_s = \rho a_y \frac{\delta y}{2}$$

$$p_z - p_s = (\rho a_z + \gamma)\frac{\delta z}{2}$$

To determine the pressure distribution at a point, we take the limit as $\delta y, \delta z \to 0$, keeping θ constant. Thus, the right-hand side of both equations goes to zero, and we get:

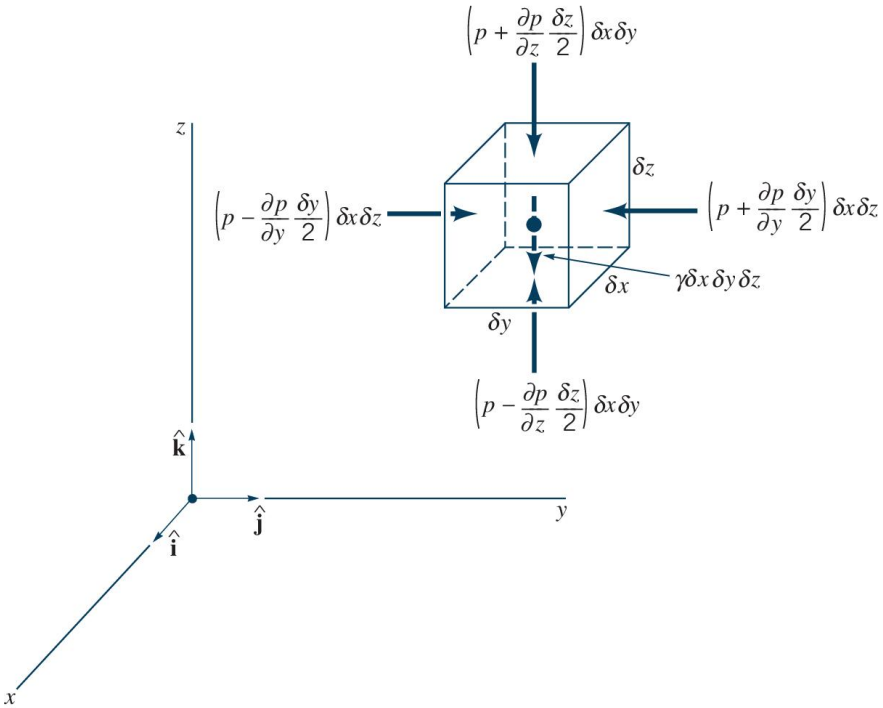
$$p_y = p_s, \quad p_z = p_s \implies p_x = p_y = p_z \quad \text{(if we had also considered } p_x\text{)}$$

Therefore, since θ is arbitrary, the pressure at a point in a fluid at rest, or in motion, is independent of direction as long as there are no shearing stresses present. This is known as **Pascal's Law**.

Note: The version of Pascal's law that states that *a change in pressure at any point in a continuous fluid is transmitted to all points in the fluid* is mainly a result of the hydrostatic pressure distribution when we neglect elevation differences across a fluid. (Same Pascal, different law?)

2.2 Basic Equation for Pressure Field

Now that we know that pressure is independent of direction at a point, we should now ask how pressure varies across points in a fluid (ie. find the pressure distribution). Consider a small rectangular fluid element with dimensions $\delta x, \delta y, \delta z$ in a fluid without shearing stresses.



For simplicity, the forces in the x direction are not shown.

There are two main forces acting on the element: surface forces due to pressure, and body force due to weight.

We define the pressure at the center of the element (x, y, z) as p. The average pressure on each face is approximated using a first-order Taylor expansion of p about the center of the element, evaluated a distance $\delta/2$ away in the corresponding coordinate direction.

The sum of surface forces in the y direction is

$$\delta F_y = \left(p - \frac{\partial p}{\partial y}\frac{\delta y}{2}\right)\delta x\,\delta z - \left(p + \frac{\partial p}{\partial y}\frac{\delta y}{2}\right)\delta x\,\delta z$$

or

$$\delta F_y = -\frac{\partial p}{\partial y}\,\delta x\,\delta y\,\delta z$$

Similarly, for the x and z directions, the sum of surface forces are

$$\delta F_x = -\frac{\partial p}{\partial x}\,\delta x\,\delta y\,\delta z, \qquad \delta F_z = -\frac{\partial p}{\partial z}\,\delta x\,\delta y\,\delta z$$

The resultant surface force acting on the element due to pressure can be expressed in vector form as

$$\delta\vec{F}_s = \delta F_x\hat{i} + \delta F_y\hat{j} + \delta F_z\hat{k}$$

or

$$\delta\vec{F}_s = -\left(\frac{\partial p}{\partial x}\hat{i} + \frac{\partial p}{\partial y}\hat{j} + \frac{\partial p}{\partial z}\hat{k}\right)\delta x\,\delta y\,\delta z$$

where $\hat{i}$, $\hat{j}$, and $\hat{k}$ are the unit vectors along the coordinate axes.

This can be rewritten in terms of the gradient operator ∇ as:

$$\frac{\delta\vec{F}_s}{\delta x\,\delta y\,\delta z} = -\nabla p, \qquad \nabla p = \frac{\partial p}{\partial x}\hat{i} + \frac{\partial p}{\partial y}\hat{j} + \frac{\partial p}{\partial z}\hat{k}$$

Combining Surface Forces with Body Forces

To describe all the forces acting on the element, we need to include the body force due to weight. This is given by:

$$-\delta\mathcal{W}\hat{k} = -\gamma\delta x\,\delta y\,\delta z\hat{k}$$

The total force acting on the element is then given by:

$$\sum\delta\vec{F} = \delta\vec{F}_s - \delta\mathcal{W}\hat{k} = \delta m\vec{a}, \quad \text{where } \delta m = \rho\delta x\,\delta y\,\delta z$$

$$\implies \boxed{-\nabla p - \gamma\hat{k} = \rho\vec{a}}$$

This is the general equation of motion for a fluid with no shearing stresses. Note that for a fluid at rest, $\vec{a} = 0$, and the equation reduces to:

$$\nabla p = -\gamma\hat{k} \implies \frac{\partial p}{\partial z} = -\gamma$$

Which once integrated, gives the hydrostatic pressure distribution:

$$p = -\gamma z + p_0$$

Where z is defined as upwards positive.

2.3 Pressure Variation at Rest

For a fluid at rest, $\vec{a} = 0$, and the equation of motion reduces to:

$$-\nabla p - \gamma \hat{k} = 0$$

$$\implies \frac{\partial p}{\partial x} = 0, \quad \frac{\partial p}{\partial y} = 0, \quad \frac{\partial p}{\partial z} = -\gamma$$

Thus, pressure only changes with depth in the z direction.

Incompressible Fluid

By definition, $\gamma = \rho g$, since changes in g is negligible for most cases, the main factor that affects γ is the density ρ. For an incompressible fluid, ρ is constant.

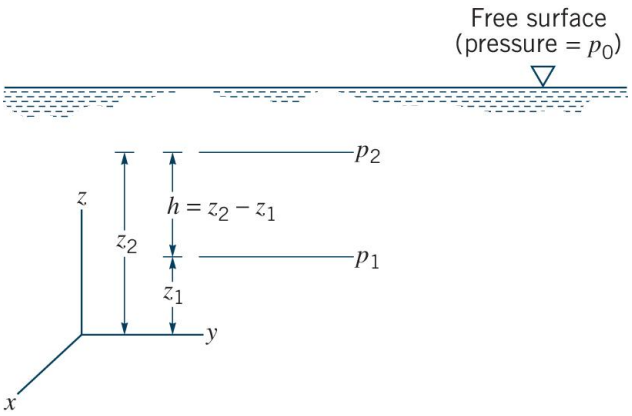
For a constant ρ, and thus γ, we can integrate the equation for pressure variation with depth:

$$\int_{p_1}^{p_2} dp = \int_{z_1}^{z_2} -\gamma dz$$

$$\implies p_2 - p_1 = -\gamma(z_2 - z_1)$$

If we let $h = z_2 - z_1$, then we can express the pressure at a depth h below a reference point as:

$$p_1 = p_2 + \gamma h$$



This is called a **hydrostatic distribution**.

Pressure Head

We can rewrite the hydrostatic distribution in terms of a height of a fluid column of specific weight γ and height h that would produce the same pressure difference:

$$h = \frac{p_1 - p_2}{\gamma}$$

Note: Often, we use the free surface of the fluid with reference pressure p_0, and thus set $p_2 = p_0$ to find pressures at depth h below the free surface.

Compressible Fluids

Compressible fluids such as gases tend to have varying density when variations in heights/elevation is very large. Thus, combining the ideal gas law ($p = \rho RT$) with the equation of motion for a fluid at rest ($\partial p/\partial z = -\gamma$), we can derive the pressure distribution for a compressible fluid.

$$\frac{dp}{dz} = -\gamma = -\rho g = -\frac{pg}{RT}$$

Integrating both sides, assuming g and R are constant,

$$\int_{p_1}^{p_2} \frac{dp}{p} = -\frac{g}{R}\int_{z_1}^{z_2} \frac{1}{T} dz$$

Before solving the integral, we must consider how temperature varies in z!

Isothermal Process

For an isothermal process, $T = T_0$ is constant, and the integral can be solved as:

$$p_2 = p_1 \exp\left[-\frac{g(z_2 - z_1)}{RT_0}\right]$$

2.4 Standard Atmosphere

The variation of temperature within the troposphere (extends to an altitude of about 11 km or $\approx$ 36,000 ft) is of the form

$$T = T_a - \beta z$$

Where

- T_a is the temperature at sea level ($z = 0$)
- β is the temperature lapse rate (rate of change of ΔT with respect to Δz). For standard atmosphere in the troposphere, $\beta = 0.0065 \frac{\text{K}}{\text{m}}$ or $0.00357 \frac{\text{R}}{\text{ft}}$

Integrating the equation for pressure variation with depth for a compressible fluid from 2.3, we have:

$$\int_{p_a}^{p} \frac{dp}{p} = -\frac{g}{R}\int_{0}^{z} \frac{1}{T} dz$$

$$p = p_a \left(1 - \frac{\beta z}{T_a}\right)^{g/(R\beta)}$$

For varying temperature in the troposphere.

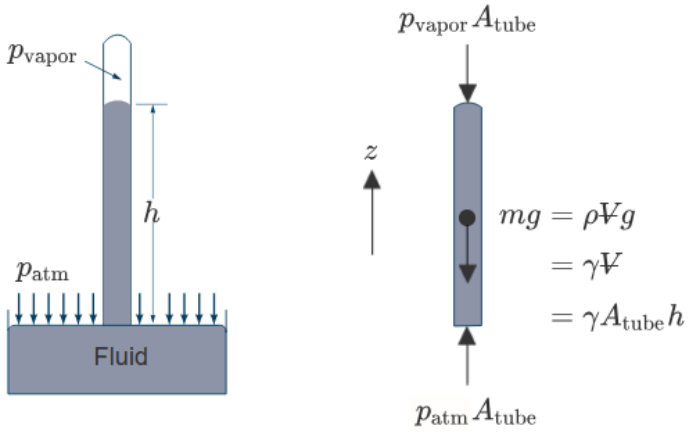
2.5 Measurement of Pressure

Mentioned before in 1.5, absolute pressure is relative to vacuum (absolute zero), gage is relative to local atmospheric pressure, and vacuum/suction is a negative gage pressure.

$$p_{abs} = p_{gage} + p_{atm}$$

Barometer

A way to measure atmospheric pressure using a column of liquid in terms of head (ex: mmHg) is by using a barometer. Consider the following barometer with a column of mercury.



For the mercury column to be at equilibrium, the pressure at the bottom of the column must be equal to the atmospheric pressure:

$$p_{atm} = \gamma h + p_{vapor}$$

Note: The vapor pressures can be found in tables 1.5 and 1.6 for different liquids at different temperatures.

This can be derived by a force balance on the mercury column:

$$\sum F_z = p_{atm} A_{tube} - \gamma A_{tube} h - p_{vapor} A_{tube} = 0$$

$$\implies p_{atm} = \gamma h + p_{vapor}$$

Assuming the vapor pressure is negligible: $p_{atm} \approx \gamma h$.

2.6 Man-O-Meters

Oh boy my favorite.

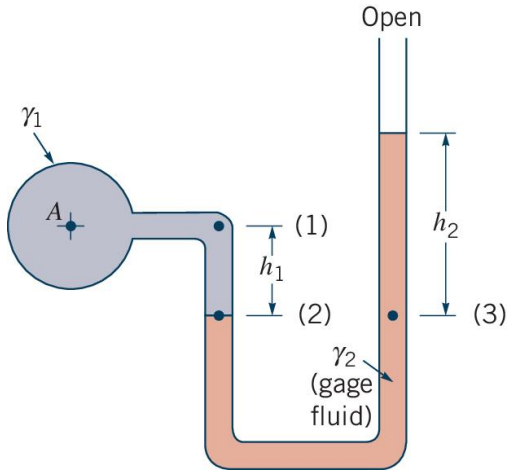
Man-O-Meters or manometers uses vertical or inclined liquid columns to measure pressure based on the same principle equation as the barometer $p = \gamma h + p_0$.

To solve manometer problems, we can use the following "rules":

1. Write pressure at either end of manometer
2. Move through manometer and add γh with increasing depth and subtract γh with decreasing depth
3. When stopping at any point, the sum is equal to the local pressure at that point

Make horizontal jumps without adding since the pressure is constant along a horizontal line.

For example, the following is a man-ohh-meter:

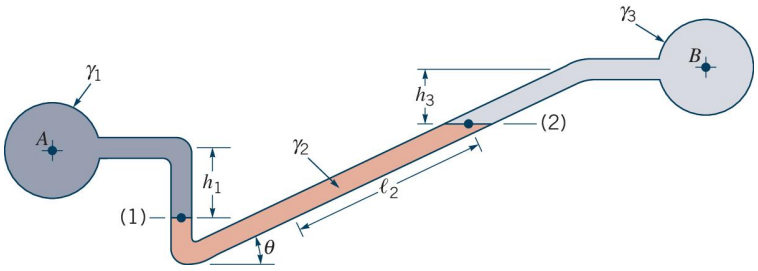


If we start from the open end:

$$p_{atm} + \gamma_2 h_2 - \gamma_1 h_1 = p_A$$

Inclined-tube Manometer

If we want to be fancy and physically add more resolution to our manometer, we can incline the tube. This allows for a longer column of liquid to measure the same pressure difference. See the following example:



The pressure difference $p_A - p_B$ can be determined by moving through the manometer starting from p_A:

$$p_A + \gamma_1 h_1 - \gamma_2 \ell_2 \sin\theta - \gamma_3 h_3 = p_B$$

$$\implies p_A - p_B = \gamma_2 \ell_2 \sin\theta + \gamma_3 h_3 - \gamma_1 h_1$$

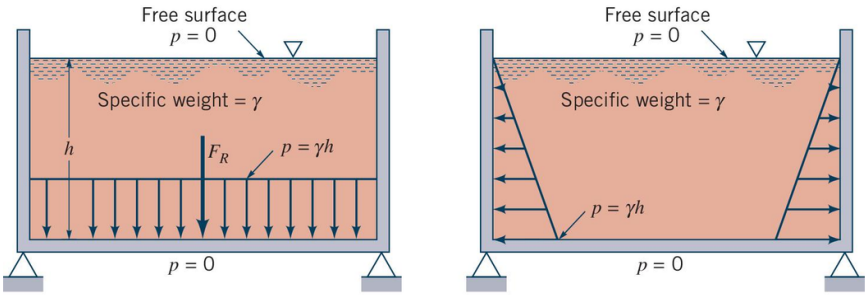
The same manometer rules applies but be careful to use the vertical height instead of the length of the inclined tube. The vertical height is given by $\ell_2 \sin\theta$.

Hydrostatics

2.8 Hydrostatic force on a plane surface

For fluid at rest, fluid pressure acts perpendicular to any surface because there is no shear stress present in a static fluid.

For example, the pressure and force distribution on an open tank is shown below:



(*a*) Pressure on tank bottom (*b*) Pressure on tank ends

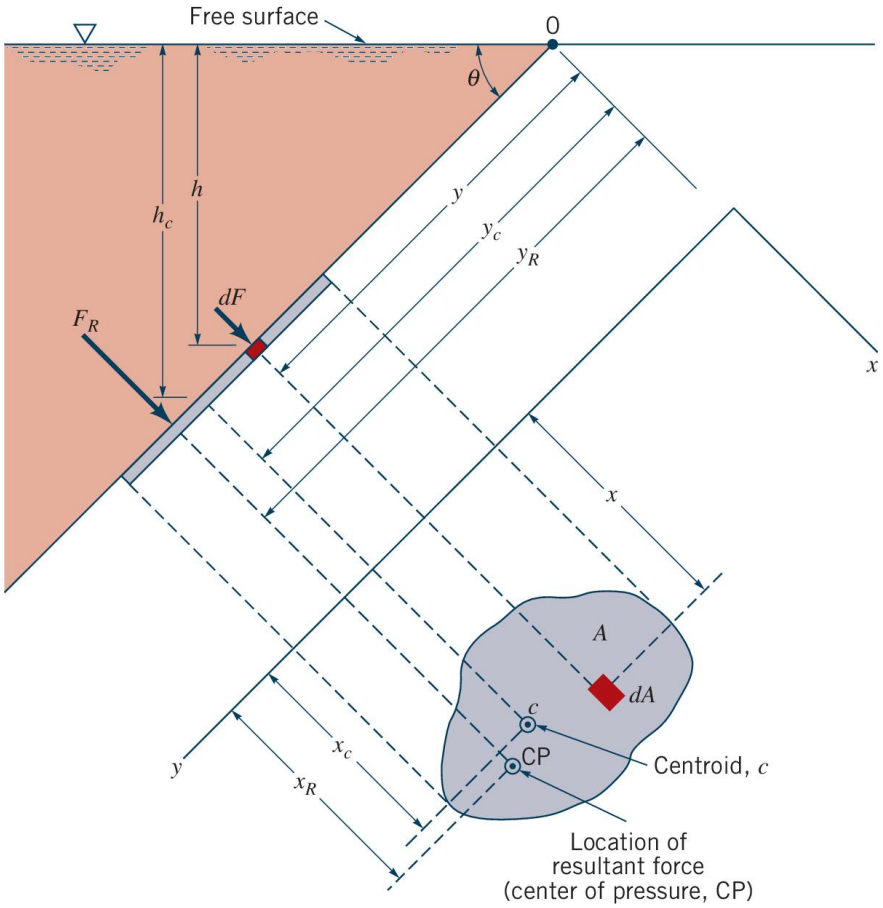
Note that if atmospheric pressure acts on both sides and bottom of the tank, it will cancel out and can be ignored (which is why we often use gage pressure). The resultant force on the bottom of tank is given by:

$$F_R = pA, \quad \text{where } p = \gamma h$$

Since the pressure is constant and uniformly distributed over the bottom, the resultant force acts through the centroid of the area. However, for the sides of the tank the pressure varies with depth, so it's harder to determine the resultant force and its line of action.

Resultant Force on a Plane Surface

Consider the following submerged inclined plane surface below. The free surface is open to the atmosphere. We define a xy coordinate system so that the surface lies on the xy plane, with the origin where the plane in which the surface lies intersect the free surface.



The force due to the fluid pressure in contact with the surface is given by integrating the pressure over the area of the surface:

$$F_R = \int_A p\, dA = \int_A \gamma h\, dA$$

Where h is the depth of the surface element dA below the free surface (h is in the vertical direction, NOT in the y direction). Note that the depth h can be expressed in terms of the y coordinate by the angle of inclination θ:

$$h = y \sin\theta$$

Also note that the integral for F_R can be expressed in terms of the first moment of area of the surface about the x-axis:

$$Q = \int_A y\, dA = A\bar{y} \quad \text{(First Moment of Area)}$$

Where y is the distance from the x-axis to the area element dA, and $\bar{y}$ is the distance from the x-axis to the centroid of the area. Thus, using some new notation, we can express the resultant force as:

$$F_R = \int_A \gamma h \, dA = \int_A \gamma y \sin\theta \, dA$$

For a constant γ and θ, and $h = y\sin\theta$:

$$F_R = \gamma \sin\theta \int_A y \, dA = \gamma \sin\theta A y_c$$

Where $\bar{y} = y_c$ is the distance from the x-axis to the centroid of the area along the y-axis. Finally:

$$\boxed{F_R = \gamma h_c A}$$

y Coordinate of the Resultant Force

Recall that the resultant force acts through the centroid of the area when the pressure is constant (like the bottom of the tank). However, when the plane surface is inclined, the pressure increases with depth, and the resultant force acts slightly below the centroid of the area.

We can determine the y coordinate of the resultant force (y_R) by summing moments about the x-axis. The moment of the resultant force at y_R is equal to the moment of the distributed pressure about the x-axis:

$$F_R y_R = \int_A y \, dF, \quad dF = \gamma h \, dA = \gamma y \sin\theta \, dA$$

$$\implies F_R y_R = \int_A \gamma \sin\theta y^2 \, dA$$

From before, $F_R = \gamma h_c A = \gamma \sin\theta A y_c$. So for a constant γ and θ:

$$\gamma \sin\theta A y_c y_R = \gamma \sin\theta \int_A y^2 \, dA$$

$$\implies y_R = \frac{\displaystyle\int_A y^2 \, dA}{y_c A}$$

Note that the integral in the numerator is the second moment of area about the x-axis, which is also known as the *area moment of inertia*:

$$I_x = \int_A y^2 \, dA$$

Thus, we can express the y coordinate of the resultant force as:

$$y_R = \frac{I_x}{y_c A}$$

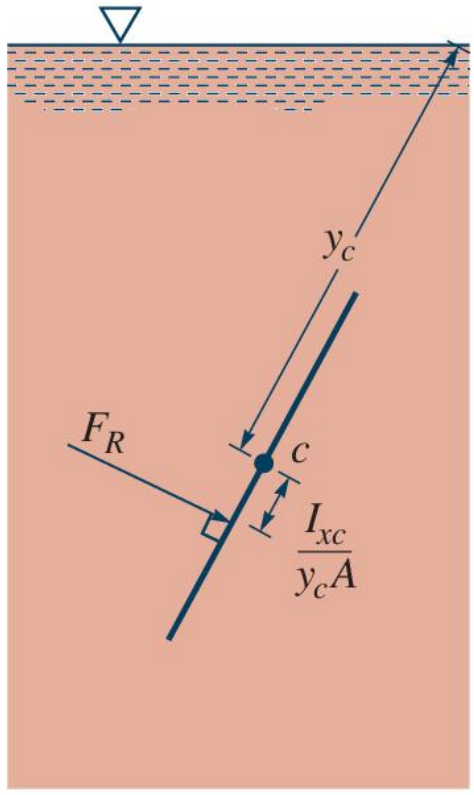
Using the parallel axis theorem, we can express I_x in terms of an axis parallel to the x-axis that passes through the centroid of the area:

$$I_x = I_{xc} + Ay_c^2$$

Thus,

$$\boxed{y_R = \frac{I_{xc}}{y_c A} + y_c}$$

Following the intuition from the beginning, since $I_{xc}/(y_c A)$ is always positive, the resultant force will always act below the centroid of the area.



Note: y_c is measured from the line of intersection of the xy plane with the surface, defining the angle θ. However, when the xy plane is horizontal and parallel to the free surface, $\theta \to 0°$, and $y_c = h_c/\sin\theta \to \infty$. In this case, the term $I_{xc}/(y_c A) \to 0$, and the resultant force acts through the centroid of the area, as expected.

x Coordinate of the Resultant Force

Similarly, we can determine the x coordinate of the resultant force (x_R) by summing moments about the y-axis. The moment of the resultant force at x_R is equal to the moment of the distributed pressure about the y-axis:

$$F_R x_R = \int_A x\, dF, \quad dF = \gamma h\, dA = \gamma y \sin\theta\, dA$$

$$\implies F_R x_R = \int_A \gamma \sin\theta x y\, dA$$

Since $F_R = \gamma h_c A = \gamma \sin\theta A y_c$, for a constant γ and θ, we can divide through by F_R to get:

$$x_R = \frac{\gamma \sin\theta \int_A xy\, dA}{\gamma \sin\theta A y_c} = \frac{I_{xy}}{y_c A}$$

Like before, I_{xy} is the product of inertia about the x and y axes. Using the parallel axis theorem, we get the general expression for the x coordinate of the resultant force:

$$\boxed{x_R = \frac{I_{xyc}}{y_c A} + x_c}$$

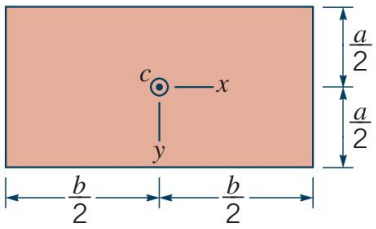
Where I_{xyc} is the product of inertia about the x and y axes that pass through the centroid of the area, and x_c is the x coordinate of the centroid of the area.

Note that if the plane surface is symmetrical about an axis parallel to either the x or y axis and passing through the centroid of the area, then $I_{xyc} = 0$, and $x_R = x_c$.

Together, x_R and y_R define the *center of pressure*, which is the point where the resultant force acts on the plane surface.

Note: For both x_R and y_R, the center of pressure will move closer to the centroid of the area as the plane surface becomes more horizontal ($\theta \downarrow$), and thus increases y_c in the denominator of both expressions. In the limit as $\theta \to 0°$, $y_c \to \infty$, and the center of pressure coincides with the centroid of the area, as expected.

Common Area Moments of Inertia



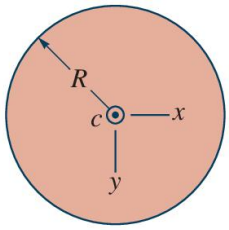
$A = ba$

$I_{xc} = \frac{1}{12} ba^3$

$I_{yc} = \frac{1}{12} ab^3$

$I_{xyc} = 0$

(*a*) Rectangle

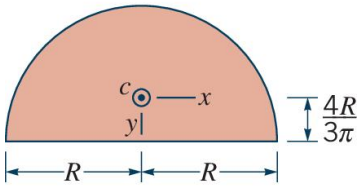


$A = \pi R^2$

$I_{xc} = I_{yc} = \frac{\pi R^4}{4}$

$I_{xyc} = 0$

(*b*) Circle



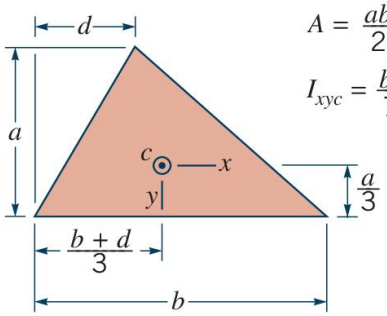
$A = \frac{\pi R^2}{2}$

$I_{xc} = 0.1098R^4$

$I_{yc} = 0.3927R^4$

$I_{xyc} = 0$

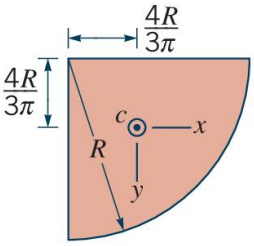
(*c*) Semicircle



$A = \frac{ab}{2}$ $I_{xc} = \frac{ba^3}{36}$

$I_{xyc} = \frac{ba^2}{72}(b - 2d)$

(*d*) Triangle



$A = \frac{\pi R^2}{4}$

$I_{xc} = I_{yc} = 0.05488R^4$

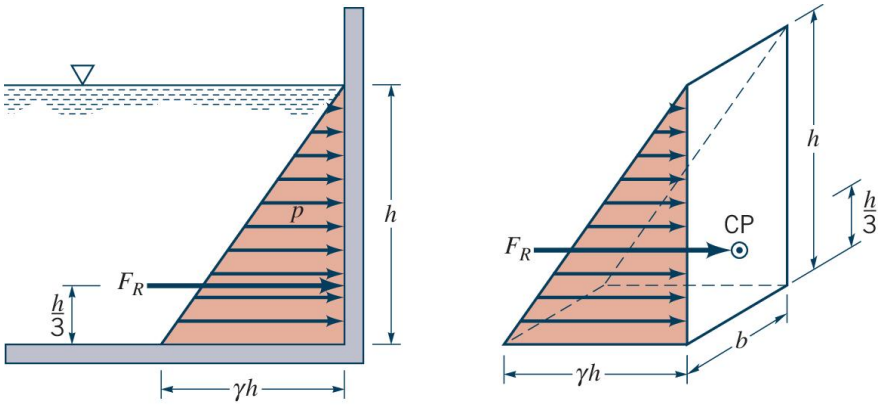
$I_{xyc} = -0.01647R^4$

(*e*) Quarter circle

2.9 Pressure Prism

This section doesn't introduce any new concepts, but rather provides a graphical approach to solving for the resultant force and center of pressure on a plane surface. This approach is often convenient for rectangular surfaces, where the centroid and volume of the pressure prism can be easily determined.

Consider the following pressure prism on a vertical plane surface with constant width b:



Since the pressure varies linearly with depth, it is zero at the free surface ($p_{min} = 0$) and maximum at the bottom of the surface ($p_{max} = \gamma h$). Thus, the average pressure on the surface is $p_{avg} = \frac{p_{min}+p_{max}}{2} = \frac{\gamma h}{2}$. The resultant force on the surface is given by:

$$F_R = p_{avg} A = \gamma \left(\frac{h}{2}\right) A$$

This is the same result as the equation $F_R = \gamma h_c A$ in section 2.8.

By forming a *pressure prism* (shown in figure (b)), the resultant force can be determined graphically as the volume of the pressure prism:

$$F_R = \text{volume} = \frac{1}{2}(\gamma h)(bh)$$

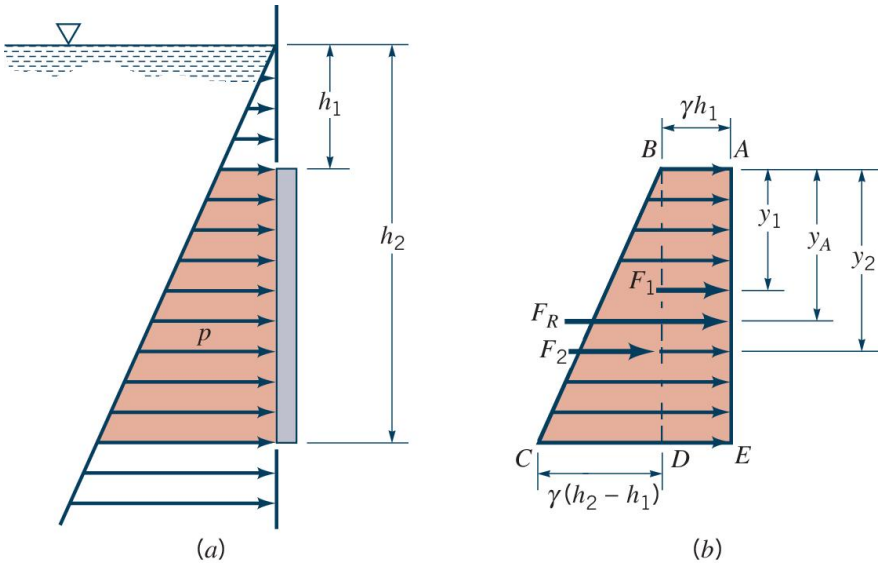
This is the same result as before as well.

The center of pressure can also be determined graphically as the centroid of the pressure prism. The centroid of a triangle is located at a distance of $\frac{1}{3}$ from the base, so the center of pressure is located at a distance of $\frac{h}{3}$ from the bottom of the surface. This is the same result as the equation $y_R = I_{xc}/(y_c A) + y_c$ in section 2.8.

Note: The pressure prism is always drawn starting from the free surface.

Rectangular Surfaces that do not intersect the free surface

Consider the following plane surface that does not intersect the free surface:



The pressure prism approach can still be used but in this case, it is trapezoidal instead of triangular. The resultant force can be found by decomposing the trapezoid into a rectangle and triangle, $ABDE$ and BCD, respectively.

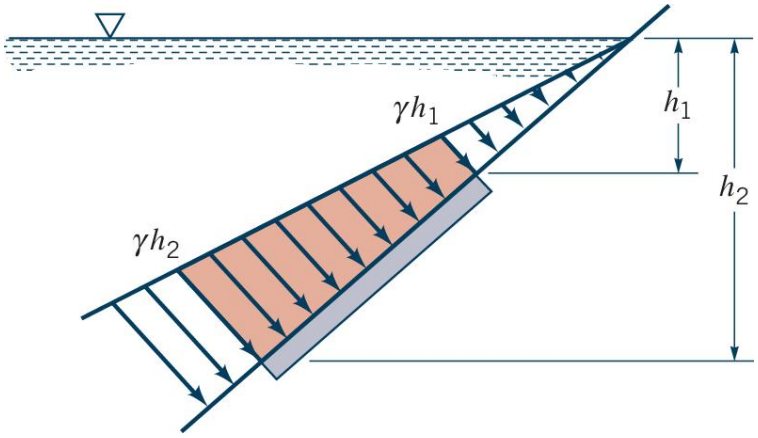
$$F_R = F_1 + F_2$$

The locations of the forces F_1 and F_2 can be found by taking the centroid of each shape, getting y_1 and y_2, respectively. The location of the resultant force can then be found by summing moments about a convenient axis (in this case, the axis passing through A):

$$F_R y_R = F_1 y_1 + F_2 y_2$$

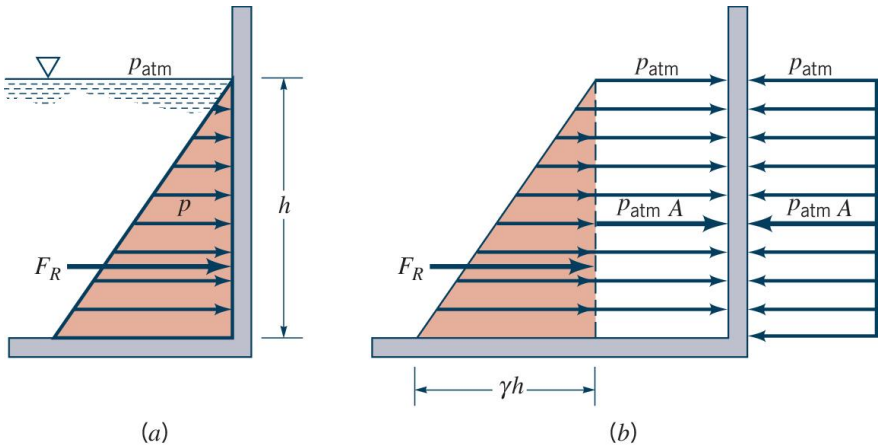
Inclined Rectangular Surfaces

A similar approach using trapezoidal pressure prisms can be used for inclined rectangular surfaces. However, note that while it is convenient to measure distances along the incline, the pressure varies with depth in the vertical direction.



Effect of Pressure at Free Surface

When we consider the effect of atmospheric pressure at the free surface, it simply offsets the pressure prism by a constant amount, resulting in a trapezoidal pressure prism.



In cases of open tanks, the atmospheric pressure acts on both sides of the vertical walls and so it cancels out. However, in cases where the pressure at the free surface is not equal to the pressure on the exterior surface, its effect is non-negligible.

Anyway, the analysis follows the same approach as the trapezoidal pressure prism.

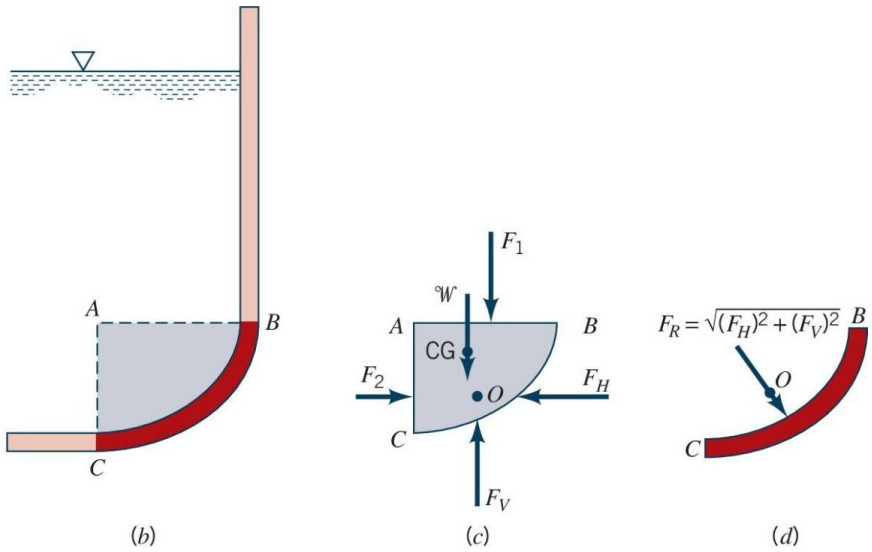
In summary, the volume of the pressure prism is basically:

$$(\text{Average pressure}) \cdot (\text{Area in contact with fluid})$$

2.10 Hydrostatic force on a curved surface

So far, we have only considered plane surfaces. What about curved surfaces? The resultant force on a curved surface can be determined by decomposing the force into horizontal and vertical components.

For example, consider the following curved surface BC:



Start by isolating a volume of fluid that is bounded by the surface of interest (b). Next, we determine the forces in the free-body diagram of the isolated fluid (c).

- F_1: Force by fluid pressure on the *planar surface* AB. This can be determined using the methods from sections 2.8 and 2.9.
- F_2: Force by fluid pressure on the *planar surface* AC. This can also be determined using the methods from sections 2.8 and 2.9.
- $\mathcal{W}$: Weight of the enclosed fluid volume. This can be determined by multiplying the volume of the fluid by its specific weight and acts through the center of gravity of the volume.
- F_H and F_V: The horizontal and vertical reaction forces from the tank on the fluid.

Since the fluid is at static equilibrium, the sum of forces in both the horizontal and vertical directions must be zero.

$$\sum F_{\text{horiz}} = 0 = F_2 - F_H \implies F_H = F_2$$

$$\sum F_{\text{vert}} = 0 = F_1 + \mathcal{W} - F_V \implies F_V = F_1 + \mathcal{W}$$

Additionally, for a body under three non-parallel forces to be in equilibrium, the lines of action of the three forces must intersect at a common point (otherwise there would be a

non-zero moment). Therefore, we can say that the resultant force F_R passes through the common intersection point of the lines of action of F_H and F_V, O.

$$F_R = \sqrt{F_H^2 + F_V^2}$$

O can be determined by summing moments to find the lines of action of F_H and F_V, and then finding their intersection.

Note: The vertical component of the resultant force can actually be solved two ways:

1. Instead of isolating a small fluid region around the surface, we can isolate the entire fluid column above it. Then, the vertical hydrostatic force on the surface is just the weight of the fluid column above it:

$$F_V = \mathcal{W} = \gamma \cancel{V}$$

2. Or alternatively, the procedure described before can be used, where we isolate a local fluid region around the surface. In that case, the vertical hydrostatic force on the surface is the sum of the weight of the fluid region and the vertical component of the hydrostatic force:

$$F_V = F_1 + \mathcal{W}$$

2.11 Buoyancy, flotation, and stability

Archimedes' Principle

The buoyant force on a body submerged in a fluid is equal to the weight of the fluid displaced by the body.

$$F_{\text{Buoyant}} = \gamma \cancel{V}$$

Line of Action for the Buoyant Force

The point through which the buoyant force acts is called the *center of buoyancy*. The buoyant force passes through the centroid of the displaced volume.

For partially submerged bodies, if the specific weight of the fluid above the liquid surface is very small compared to the displaced fluid, the center of buoyancy can be approximated as the centroid of the submerged volume. This is a common approximation when the fluid above the surface is air.

If a body is submerged in a fluid of varying specific weight, such as a layered fluid, the magnitude of the buoyant force remains equal to the weight of the displaced fluid. However, the center of buoyancy is located at the center of gravity of the displaced fluid.

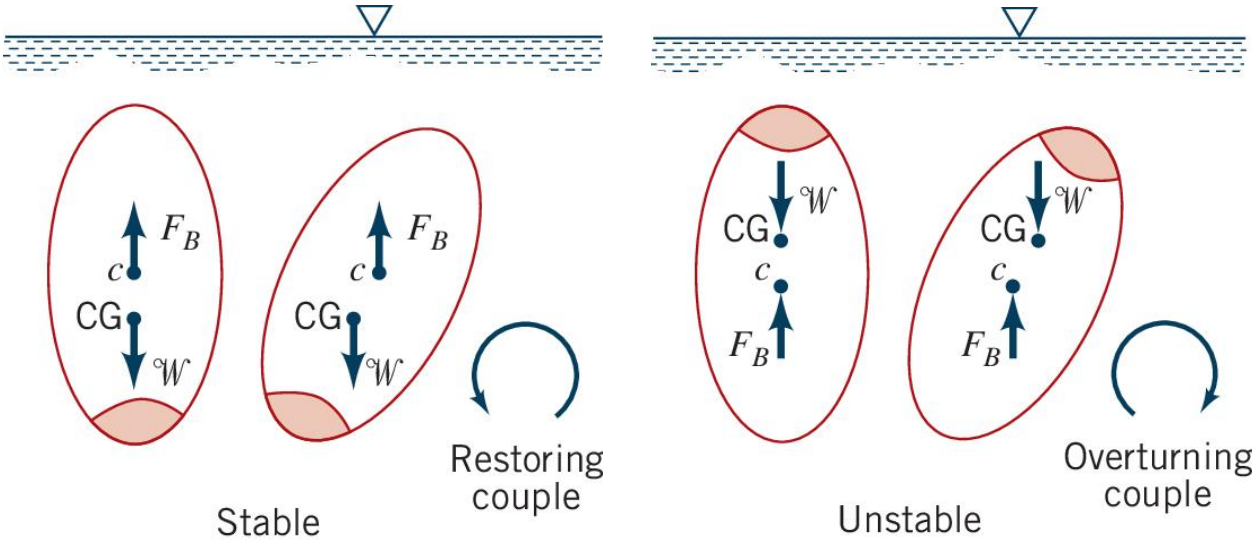
Note: The buoyant force is a special application of the hydrostatic pressure forces acting on the surfaces of a body. In other words, both buoyancy and hydrostatic pressure forces arise from the same physical phenomenon. Therefore, when performing calculations, use

either the resultant buoyant force or the individual hydrostatic pressure forces, but do not include both.

Stability

A body is said to be in stable equilibrium if it returns to its original position after being slightly disturbed, and in unstable equilibrium if it moves to a new equilibrium position instead.

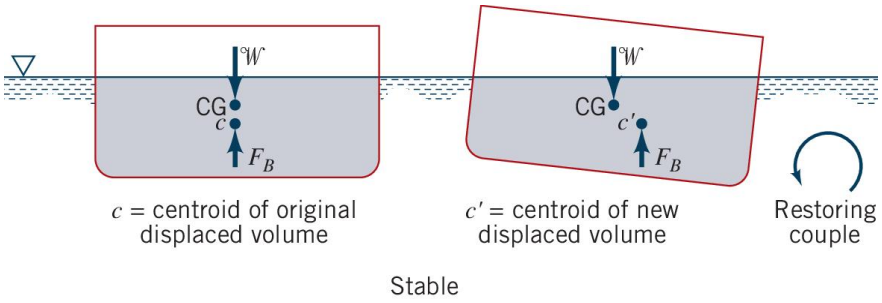
Stability of a submerged body can be determined by comparing the center of gravity of the body and the center of buoyancy:



The body is in stable equilibrium if the center of gravity is below the center of buoyancy, and in unstable equilibrium if the center of gravity is above the center of buoyancy.

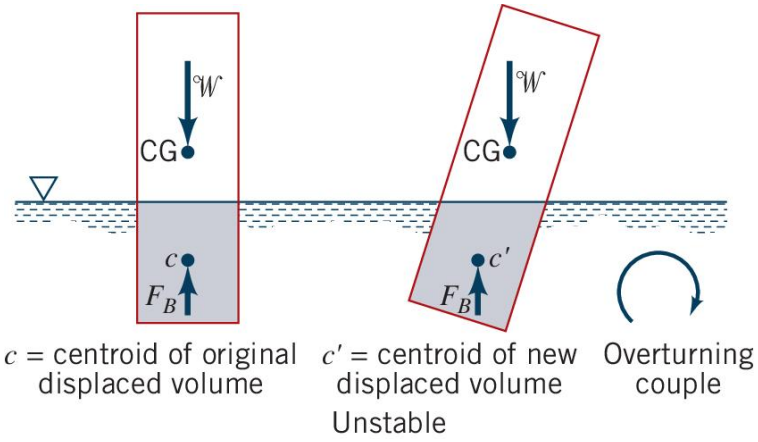
Floating Body Stability

For a floating body, the center of buoyancy (which is approximately the centroid of the submerged volume) *moves as the body is tilted*. Thus, it is possible for a wide floating body to be in stable equilibrium even if its center of gravity is above the center of buoyancy as seen below:



However, a narrow floating body may be in unstable equilibrium if its center of gravity

is above the center of buoyancy since the center of buoyancy may not move enough to counteract the tilt.



2.12 Pressure variation in a fluid with rigid-body motion

Recall that the general equation of motion for a fluid with no shearing stresses is given by:

$$-\nabla p - \gamma \hat{k} = \rho \vec{a}$$

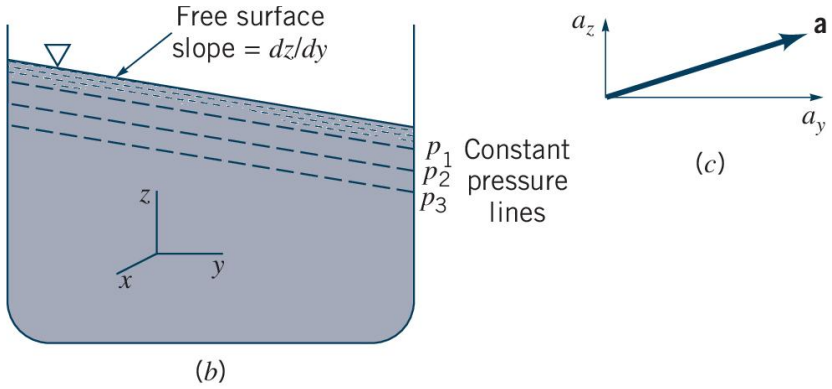
$$\implies -\frac{\partial p}{\partial x} = \rho a_x, \quad -\frac{\partial p}{\partial y} = \rho a_y, \quad -\frac{\partial p}{\partial z} = \gamma + \rho a_z$$

Problems involving fluids in motion but having no shearing stresses occur when the fluid undergoes rigid-body motion. Under rigid-body motion, fluid particles maintain fixed positions relative to one another, so there is no deformation/shearing (after some transient sloshing dies down). For example, if a container of fluid accelerates along a straight path, the fluid will move as a rigid mass, and the fluid would not deform (i.e., no shearing stresses).

By using the equation of motion for a fluid with no shearing stresses, we can determine the pressure distribution in a fluid in motion.

Linear Motion

Consider an open container of fluid that is moving with a constant acceleration a along a straight path.



Since $a_x = 0$ it implies that $\partial p/\partial x = 0$, and thus the pressure is constant along the x direction. Additionally, in the y and z directions, the pressure distribution is given by:

$$\frac{\partial p}{\partial y} = -\rho a_y$$

$$\frac{\partial p}{\partial z} = -\gamma - \rho a_z = -\rho(g + a_z)$$

The change in pressure between two closely spaced points in the fluid at (x, y, z) and $(x, y + dy, z + dz)$ is given by the first-order Taylor series expansion:

$$dp = \frac{\partial p}{\partial y}dy + \frac{\partial p}{\partial z}dz$$

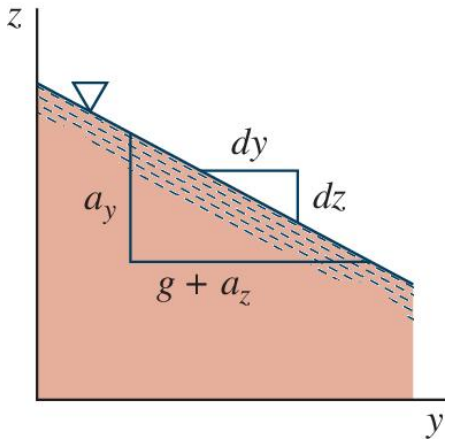
Plugging in the expressions for $\partial p/\partial y$ and $\partial p/\partial z$, we get:

$$dp = -\rho a_y dy - \rho(g + a_z)dz$$

Along a line of constant pressure, $dp = 0$ by definition, thus the equation above can be rearranged to give the slope of the constant pressure line:

$$\frac{dz}{dy} = -\frac{a_y}{g + a_z}$$

Note that for any fluid with a nonzero horizontal acceleration component, $a_y \neq 0$ and thus $dz/dy \neq 0$, which means that the free surface of the fluid will have a non-zero slope. Additonally, ALL lines of constant pressure will be parallel to the free surface (with the same slope).



Vertical Acceleration

For the special case of pure vertical acceleration where $a_y = 0$ and $a_z \neq 0$, the slope of the free surface is zero, and the free surface remains horizontal. However, the pressure distribution is NOT hydrostatic but is given by:

$$\frac{dp}{dz} = -\rho(g + a_z)$$

In other words, the pressure distribution is still linear but now includes a term for the vertical acceleration:

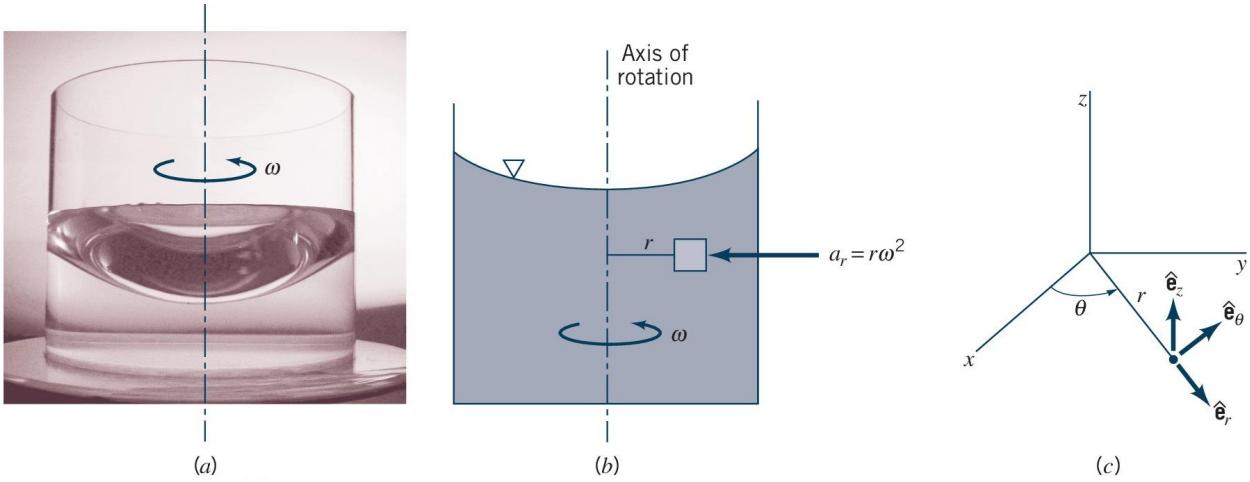
$$p = p_0 - \rho(g + a_z)z$$

Note: Pressure still increases with depth, but here z is defined as positive upwards. So, $z < 0$ below the free surface.

It can also be seen that if the fluid is in free fall, $a_z = -g$, then $dp/dz = 0$, and the pressure is constant throughout the fluid.

Rigid-body rotation

Consider a container of fluid that is rotating about a vertical axis with a constant angular velocity ω.



Since the paths of all fluid particles are circular, the acceleration of a fluid particle a distance r from the axis of rotation is given by $r\omega^2$, and it is more convenient to use cylindrical coordinates (r, θ, z) instead of Cartesian coordinates (x, y, z). The gradient operator in cylindrical coordinates is given by:

$$\nabla = \frac{\partial}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial}{\partial \theta}\hat{e}_\theta + \frac{\partial}{\partial z}\hat{e}_z$$

Thus the equation of motion for a fluid with no shearing stresses in cylindrical coordinates is given by:

$$\begin{aligned} -\nabla p - \gamma \hat{e}_z &= \rho \vec{a} \\ -\frac{\partial p}{\partial r}\hat{e}_r - \frac{1}{r}\frac{\partial p}{\partial \theta}\hat{e}_\theta - \left(\frac{\partial p}{\partial z} + \gamma\right)\hat{e}_z &= \rho a_r \hat{e}_r + \rho a_\theta \hat{e}_\theta + \rho a_z \hat{e}_z \end{aligned}$$

Since the accelerations in the cylindrical coordinate system are given by:

$$\vec{a}_r = -r\omega^2\,\hat{e}_r, \quad \vec{a}_\theta = 0, \quad \vec{a}_z = 0$$

Thus, setting components equal:

$$\frac{\partial p}{\partial r} = \rho r \omega^2, \quad \frac{1}{r}\frac{\partial p}{\partial \theta} = 0, \quad \frac{\partial p}{\partial z} = -\gamma$$

Since there are two non-zero components, the pressure distribution is in terms of two variables, r and z. Again with the first order Taylor series expansion, the change in pressure between two closely spaced points in the fluid at (r, θ, z) and $(r + dr, \theta, d\theta, z + dz)$ is given by:

$$dp = \frac{\partial p}{\partial r}dr + \frac{\partial p}{\partial z}dz \quad \Longrightarrow \quad dp = \rho r \omega^2 dr - \gamma dz$$

For a surface of constant pressure, $dp = 0$, and using $\gamma = \rho g$, the slope of the constant pressure surface is given by:

$$\frac{dz}{dr} = \frac{r\omega^2}{g}$$

Integrating this differential equation gives the equation of the free surface of the fluid as a parabolic surface:

$$z(r) = \frac{\omega^2}{2g}r^2 + \text{constant}$$

Pressure Distribution

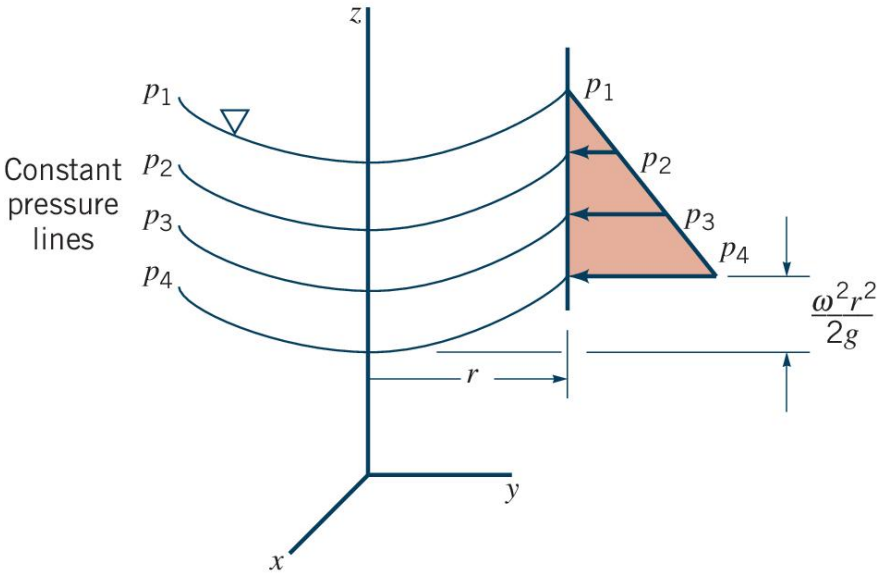
Integrating the equation for dp gives the pressure distribution in the fluid:

$$\int dp = \int \rho r \omega^2 dr - \int \gamma dz$$

$$\Longrightarrow \; p(r, z) = \frac{1}{2}\rho\omega^2 r^2 - \gamma z + \text{constant}$$

These relations show that

- The free surface of a fluid in rigid-body rotation is parabolic.
- While the pressure distribution of a fluid in rigid-body rotation depends on both r and z, at a fixed radius r, the pressure distribution is linear with depth z.



Note: Once again note that positive z is defined as upward, so z is negative below the free surface.

Also note that when calculating the pressure *difference*, the constant of integration cancels out. Otherwise, the constant of integration can be determined by applying a boundary condition, such as the pressure at the free surface.

Elementary Fluid Dynamics—The Bernoulli Equation

3.1 Newton's Second Law

The Bernoulli equation is one of the most used and abused equations in fluid mechanics. It is derived from Newton's second law on a fluid particle as it behaves in an "ideal" manner. Note that we must respect the assumptions of the Bernoulli equation, otherwise it is not valid.

Let's define a **fluid particle** to be a small volume of fluid that is large enough to contain a sufficient number of molecules. As the fluid particle moves, it usually experiences an acceleration and thus the net force on the particle is non-zero: $\vec{F} = m\vec{a}$.

We will consider fluid motion while neglecting frictional effects due to shear stresses in a flowing fluid. To do this, we will consider a fluid that has no viscosity, called an **inviscid**

fluid. The resulting flow is called **inviscid flow**.

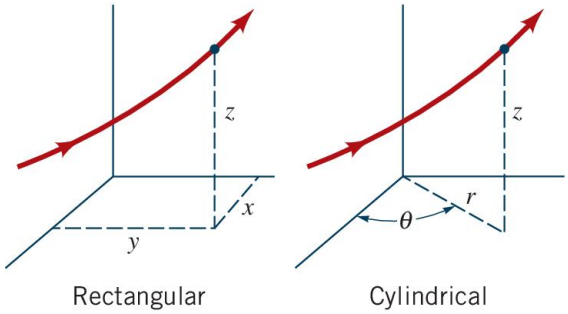
In reality there are no inviscid fluids, since every fluid experiences shear stresses due to viscosity. However, in many situations, the viscous effects are relatively small compared to other effects (such as inertia and pressure). Also note that the flow of gas is usually inviscid, since the viscosity of gases is very small. But always check whether the viscous effects are negligible before applying the Bernoulli equation.

For an inviscid fluid, the only forces acting on a fluid particle are body forces (gravity) and surface forces (pressure). So the net force on a fluid particle is given by:

(Net pressure force on particle)+(Net gravity on particle) = (Particle mass)·(Particle acceleration)

Coordinate System

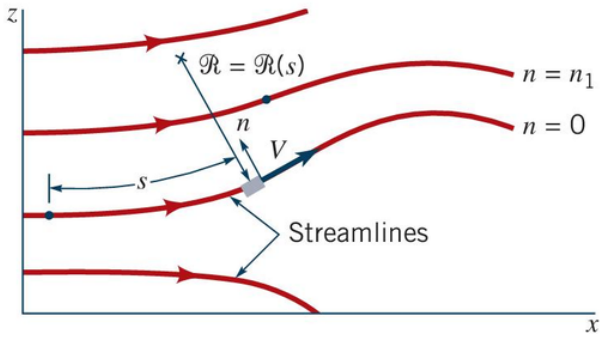
The rectangular and cylindrical coordinate systems are commonly used.



For **steady flow** (where there are no time dependences), fluid particles follow fixed **streamlines**. In other words, successive particles that pass through the same point in space will follow the same fixed path. Thus, the velocity vector of a fluid particle is always tangent to the streamline it is on.

It is then more convenient to use a **streamline coordinate system** in terms of:

- $s = s(t)$: the distance along the streamline from some reference point
- n: the perpendicular distance from a reference streamline
- $\mathscr{R} = \mathcal{R}(s)$: the local radius of curvature of the streamline at s



Velocity and Acceleration

In terms of streamline coordinates, the velocity of a fluid particle is given by:

$$\vec{V} = \frac{ds}{dt}\hat{e}_s$$

Where $\hat{e}_s$ is the unit vector tangent to the streamline.

The acceleration of a fluid particle in 2D flow would have two components: along the streamline a_s and perpendicular to the streamline a_n. By definition, the streamwise acceleration is given by:

$$a_s = \frac{dV}{dt}$$

Generally, velocity can depend on both position and time $V = V(s, n, t)$. The multivariable chain rule gives:

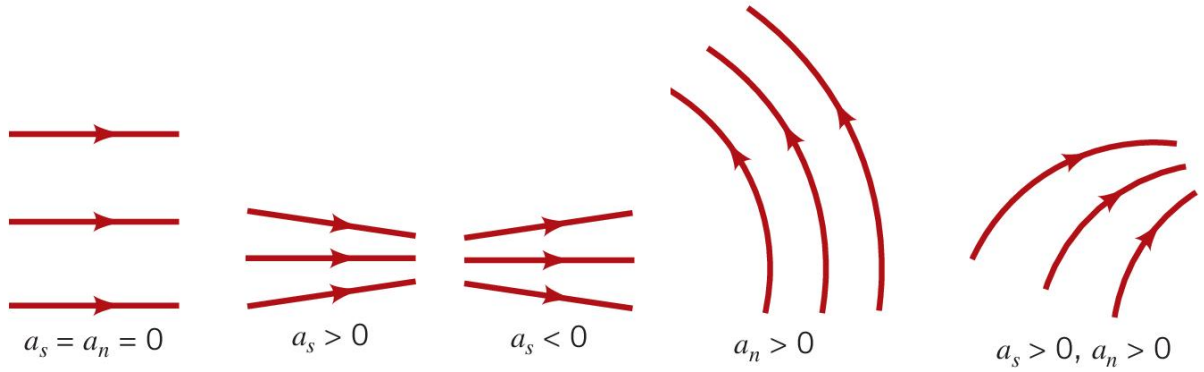
$$\frac{dV}{dt} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial s}\frac{ds}{dt} + \frac{\partial V}{\partial n}\frac{dn}{dt}$$

However, for steady flow $\partial V/\partial t = 0$, and since we are looking at the acceleration along the streamline, $dn/dt = 0$ since the particle is not moving between streamlines. Thus, the streamwise acceleration is given by:

$$\boxed{a_s = V\frac{\partial V}{\partial s}}$$

The normal acceleration is given by the centripetal acceleration of the particle as it moves along a curved path:

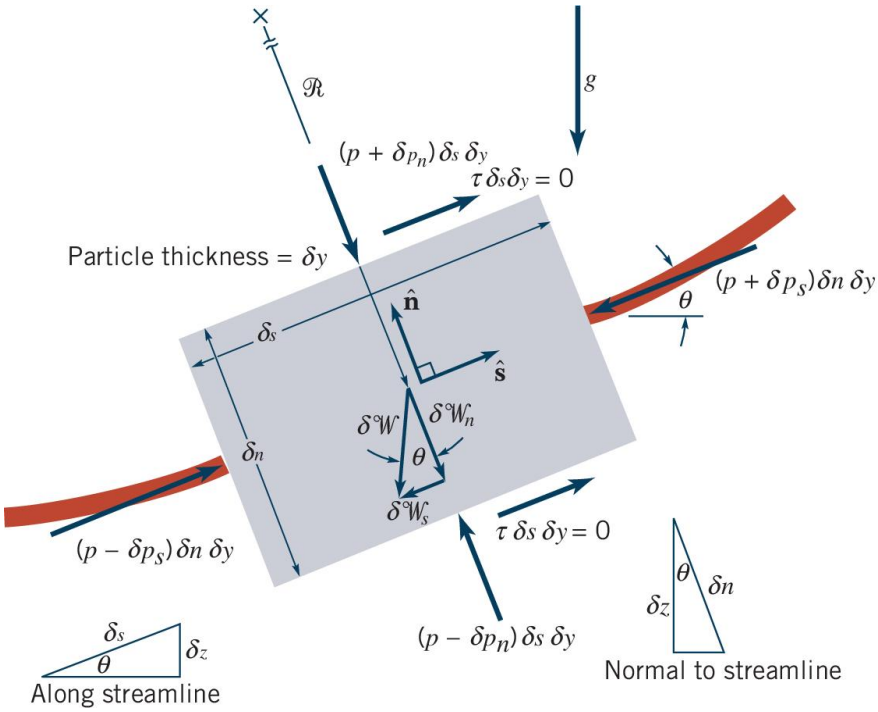
$$\boxed{a_n = \frac{V^2}{\mathscr{R}}}$$



Intuitively, when streamlines converge, velocity increases (by conservation of mass), and thus the streamwise acceleration is positive. When streamlines diverge, velocity decreases, and thus the streamwise acceleration is negative.

3.2 $F = ma$ along a streamline (Bernoulli Equation)

Consider a small fluid particle of size δs by δn along a streamline and δy normal to the plane of the streamline.



Newton's second law along the streamline gives:

$$\sum \delta F_s = \delta m\, a_s = \delta m\, V\frac{\partial V}{\partial s} = \rho \delta \cancel{\mathcal{V}}\, V\frac{\partial V}{\partial s}$$

Where $\delta \cancel{\mathcal{V}} = \delta s \delta n \delta y$ is the volume of the fluid particle. Note that this equation is valid for both compressible and incompressible fluids since the density ρ is not assumed to be constant.

Sum of Forces along a streamline

For inviscid flow, the only forces acting on a fluid particle are body forces (gravity) and surface forces (pressure). Shear forces are zero ($\tau \delta s \delta y = 0$).

The weight of the fluid particle is given by:

$$\delta \mathcal{W} = \rho g \delta \cancel{\mathcal{V}} = \gamma \delta \cancel{\mathcal{V}}$$

$$\implies \delta \mathcal{W}_s = -\delta \mathcal{W} \sin\theta = -\gamma \delta \cancel{\mathcal{V}} \sin\theta$$

Where θ is the angle between the streamline and the horizontal.

Suppose the pressure at the center of the particle is $p(s, n) = p$. Thus, the pressure at

the front and back of the particle (located at $s \pm \delta s/2$) are given by a small change in pressure δp_s: $p(s \pm \delta s/2, n) = p \pm \delta p_s$:

$$p(s + \Delta s) \approx p(s) + \frac{\partial p}{\partial s}\Delta s \quad \text{(first order taylor series approx)}$$

$$\implies \delta p_s = \frac{\partial p}{\partial s}\frac{\delta s}{2}$$

Thus the net pressure force on the particle along the streamline is given by:

$$\delta F_{ps} = (p - \delta p_s)\delta n \delta y - (p + \delta p_s)\delta n \delta y = -2\delta p_s \delta n \delta y = -\frac{\partial p}{\partial s}\delta s \delta n \delta y$$

$$\implies \delta F_{ps} = -\frac{\partial p}{\partial s}\delta \cancel{\forall}$$

Note that it is the pressure gradient, not the pressure itself, that produces a net force on the fluid particle. $\nabla p = \partial p/\partial s\, \hat{e}_s + \partial p/\partial n\, \hat{e}_n$

- $\partial p/\partial s > 0$: pressure increases along the streamline, producing a force opposite to the direction of flow
- $\partial p/\partial s < 0$: pressure decreases along the streamline, producing a force in the direction of flow

Combining both the gravity and pressure forces along the streamline, we get:

$$\sum \delta F_s = \delta \mathcal{W}_s + \delta F_{ps} = \left(-\gamma \sin\theta - \frac{\partial p}{\partial s}\right)\delta \cancel{\forall}$$

Setting the sum of forces equal to the mass times acceleration, we get:

$$\left(-\gamma \sin\theta - \frac{\partial p}{\partial s}\right)\delta \cancel{\forall} = \rho \delta \cancel{\forall}\, V \frac{\partial V}{\partial s}$$

$$\boxed{-\gamma \sin\theta - \frac{\partial p}{\partial s} = \rho\, V \frac{\partial V}{\partial s}}$$

Recall that $a_s = V\partial V/\partial s$ is the streamwise acceleration of the fluid particle.

Derivation of the Bernoulli Equation

Starting from the $F = ma$ equation along a streamline, we want to rewrite the equation in terms of differentials so that we can integrate it along a streamline ds.

$$-\gamma \sin\theta - \frac{\partial p}{\partial s} = \rho\, V \frac{\partial V}{\partial s}$$

We can do the following:

- From the FBD before where z is the vertical axis.

$$\sin\theta = \frac{dz}{ds}$$

- From the total differential of pressure $p(s, n)$:

$$dp = \frac{\partial p}{\partial s} ds + \frac{\partial p}{\partial n} dn$$

  - Recall that the multivariable chain rule gives:

$$\frac{dp}{dt} = \frac{\partial p}{\partial s}\frac{ds}{dt} + \frac{\partial p}{\partial n}\frac{dn}{dt}$$

  - Multiplying both sides by dt gives:

$$dp = \frac{\partial p}{\partial s} ds + \frac{\partial p}{\partial n} dn$$

$$\frac{ds}{dt} dt = ds, \quad \frac{dn}{dt} dt = dn$$

  Along a streamline n is constant, so $dn = 0$, and thus

$$dp = \frac{\partial p}{\partial s} ds \implies \frac{dp}{ds} = \frac{\partial p}{\partial s}$$

- Similarly, velocity is a multivariable function $v(s, n)$ and along a streamline n is constant, so we can write:

$$\frac{dV}{dt} = \frac{\partial V}{\partial s}\frac{ds}{dt} + \frac{\partial V}{\partial n}\frac{dn}{dt}$$

  Since $dn = 0$ along a streamline, we get:

$$\frac{dV}{ds} = \frac{\partial V}{\partial s}$$

  Thus, we can rewrite the streamwise acceleration as:

$$a_s = V\frac{\partial V}{\partial s} = V\frac{dV}{ds} = \frac{1}{2}\frac{d}{ds}(V^2)$$

Differential form of the Bernoulli Equation

Putting it all together, we can write the differential form of the Bernoulli Equation as:

$$-\gamma\frac{dz}{ds} - \frac{dp}{ds} = \frac{1}{2}\rho\frac{d(V^2)}{ds}$$

Multiplying through by ds gives:

$$\boxed{dp + \frac{1}{2}\rho d(V^2) + \gamma dz = 0}$$

Note: To get to this point we had to make the following assumptions:

- Net forces: no shear stresses (inviscid flow), only pressure and gravity forces
- Streamwise acceleration: steady flow $\partial V/\partial t = 0$
- Along a streamline: $dn = 0$ (no motion between streamlines)

Integral form of the Bernoulli Equation
Divide the differential form by ρ, the indefinite integral gives:

$$\int \frac{dp}{\rho} + \int \frac{1}{2} d(V^2) + \int g dz = C$$

$$\boxed{\int \frac{dp}{\rho} + \frac{1}{2} V^2 + gz = C}$$

We generally can't get rid of the integral $\int dp/\rho$ since ρ is not a constant. For example, for an ideal gas, $\rho = p/RT$.

However, for an incompressible fluid, ρ is constant, and thus we can write:

$$\boxed{p + \frac{1}{2}\rho V^2 + \gamma z = \text{constant along a streamline}}$$

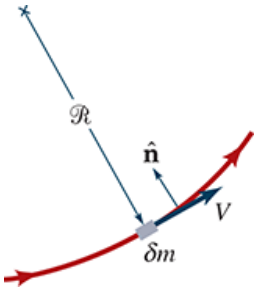
3.3 $F = ma$ normal to a streamline

In section 3.2, we looked at how forces along a streamline affects a fluid particle's speed. In this section, we will apply the same Newton's second law approach to explain how forces normal to a streamline *changes the direction of a fluid particle's velocity.*

Recall from section 3.1 that the normal acceleration of a fluid particle for steady flow is given by:

$$a_n = \frac{V^2}{\mathscr{R}}$$

with a direction towards the center of curvature of the streamline.

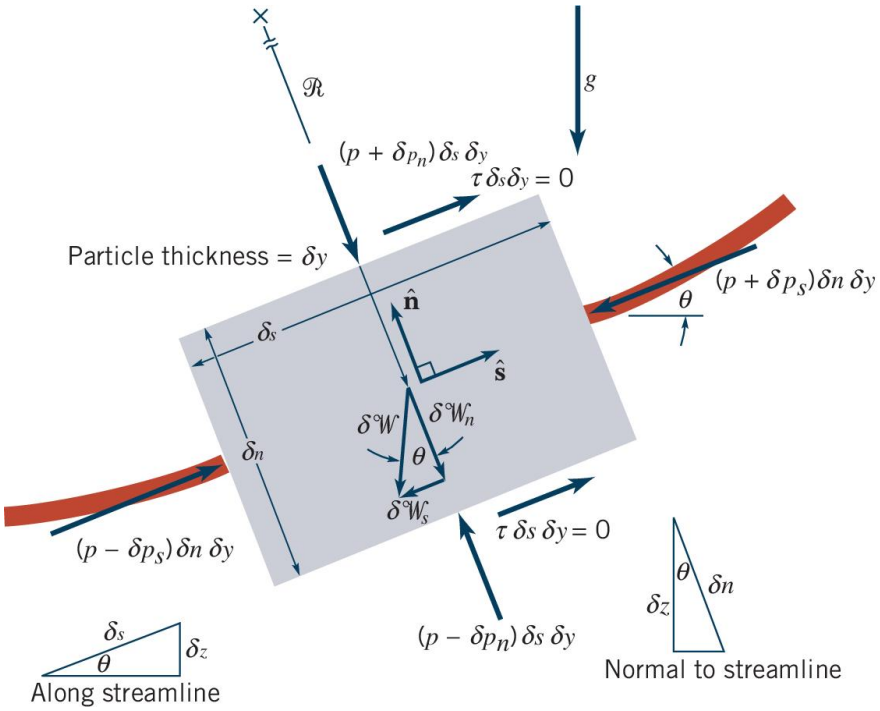


Then Newton says:

$$\sum \delta F_n = \delta m \frac{V^2}{\mathscr{R}} = \rho \delta \forall \frac{V^2}{\mathscr{R}}$$

Sum of Forces normal to a streamline

Consider the same FBD of a fluid particle as in section 3.2:



Again, assuming inviscid flow, the only forces acting on the fluid particle are gravity and pressure forces.

The component of the weight in the normal direction is given by:

$$\delta\mathcal{W}_n = -\delta\mathcal{W}\cos\theta = -\gamma\delta\cancel{\mathcal{V}}\cos\theta$$

Similarly to the previous section, letting the pressure at the center of the particle be $p(s,n) = p$, the pressure at the top and bottom of the particle (located at $n \pm \delta n/2$) are given by a small change in pressure δp_n: $p(s, n \pm \delta n/2) = p \pm \delta p_n$:

$$p(s, n + \Delta n) \approx p(s,n) + \frac{\partial p}{\partial n}\Delta n \quad \text{(first order taylor series approx)}$$

$$\implies \delta p_n = \frac{\partial p}{\partial n}\frac{\delta n}{2}$$

Thus the net pressure force on the particle in the normal direction is given by:

$$\delta F_{pn} = (p - \delta p_n)\delta s\delta y - (p + \delta p_n)\delta s\delta y = -2\delta p_n\delta s\delta y = -\frac{\partial p}{\partial n}\delta s\delta n\delta y$$

$$\delta F_{pn} = -\frac{\partial p}{\partial n}\delta\cancel{\mathcal{V}}$$

The sum of forces in the normal direction is then given by:

$$\sum \delta F_n = \delta\mathcal{W}_n + \delta F_{pn} = \left(-\gamma\cos\theta - \frac{\partial p}{\partial n}\right)\delta\cancel{\mathcal{V}}$$

Setting the sum of forces equal to the mass times acceleration, we get:

$$\left(-\gamma\cos\theta - \frac{\partial p}{\partial n}\right)\delta\cancel{\mathscr{V}} = \rho\delta\cancel{\mathscr{V}}\frac{V^2}{\mathscr{R}}$$

Canceling $\delta\cancel{\mathscr{V}}$ and substituting $\cos\theta = dz/dn$ gives:

$$\boxed{-\gamma\frac{dz}{dn} - \frac{\partial p}{\partial n} = \rho\frac{V^2}{\mathscr{R}}}$$

This is the **normal equation of motion** for steady inviscid flow. It shows the pressure gradient and gravitational force that is required to keep a fluid particle moving along a curved streamline.

Special Case

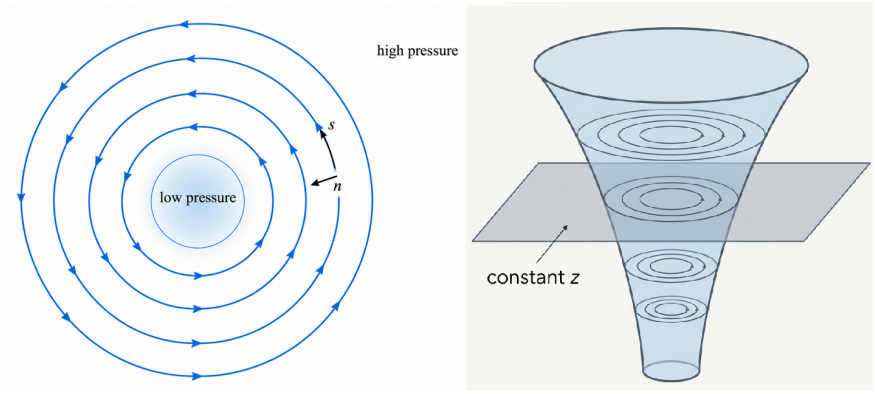
In the case where gravity is negligible compared to pressure forces (often the case for gases), $\gamma \approx 0$. Alternatively, if the flow is horizontal, $dz/dn = 0$, and gravity has no component normal to the streamline. In either case, the normal equation of motion reduces to:

$$\frac{\partial p}{\partial n} = -\rho\frac{V^2}{\mathscr{R}}$$

Horizontal Flow

Horizontal flow is interesting because it explains the pressure distribution required to sustain curved fluid motion, such as in vortexes.

Consider a top-down view of the fluid motion of a vortex at a constant height z. Because V, ρ, and $\mathscr{R}$ are all positive, this means that $\partial p/\partial n < 0$, and thus the pressure decreases as n increases (as we move towards the center of curvature). This is why the pressure inside a tornado tends to be lower than the pressure outside the tornado.



Integrating across streamlines

Like before in section 3.2 we can integrate the Newton's second law equation by writing it in terms of differentials. Starting from:

$$-\gamma\frac{dz}{dn} - \frac{\partial p}{\partial n} = \rho\frac{V^2}{\mathscr{R}}$$

Using the total differential of pressure, now with s constant ($ds = 0$), we can say that

$$\frac{\partial p}{\partial n} = \frac{dp}{dn}$$

Thus, we can write the normal equation of motion in terms of differentials as:

$$-\gamma dz - dp = \rho \frac{V^2}{\mathscr{R}} dn$$

The indefinite integral gives:

$$\boxed{\int \frac{dp}{\rho} + \int \frac{V^2}{\mathscr{R}} dn + gz = \text{constant across the streamline}}$$

Note: To get to this point we had to make the following assumptions:

- Net forces: no shear stresses (inviscid flow), only pressure and gravity forces
- Normal acceleration: steady flow $\partial V/\partial t = 0$
- Across a streamline: $ds = 0$ (only motion between streamlines)

For an incompressible fluid, ρ is constant, and thus we can write:

$$\boxed{p + \rho \int \frac{V^2}{\mathscr{R}} dn + \gamma z = \text{constant across the streamline}}$$

We cannot get rid of the integral $\int V^2/\mathscr{R} dn$ since V and $\mathscr{R}$ are not constant across streamlines.

Duck:

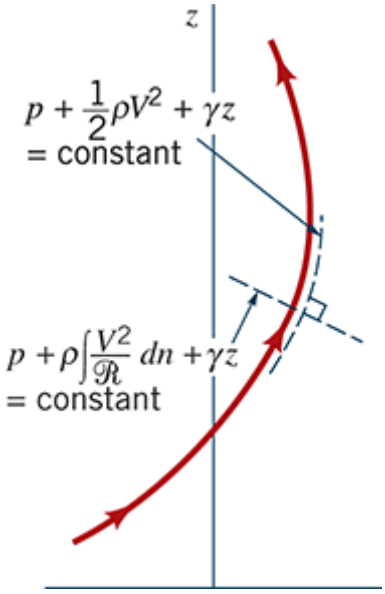


3.4 Physical interpretations and alternate forms of the Bernoulli equation

In sections 3.2 and 3.3, we have derived the basic governing equations for inviscid flow along and across streamlines. Both require steady, inviscid, incompressible flow

$p+\frac{1}{2}\rho V^2+\gamma z =$ constant along the streamline

$p+\rho\int \frac{V^2}{\mathscr{R}}\,dn+\gamma z =$ constant across the streamline



Bernoulli's equation as the work-energy principle

Recall the work-energy principle:

$$W_{\text{net}} = \Delta KE$$

The work done on a mass by a net force is equal to the change in kinetic energy of the mass.

When we integrated the Newton's second law equation along a streamline, we essentially found the net work since $W = \int \vec{F}\cdot d\vec{s}$. The work done by the pressure and gravity forces on a fluid particle is equal to the change in kinetic energy of the fluid particle.

$$p+\frac{1}{2}\rho V^2+\gamma z = \text{constant along the streamline}$$

Dividing through by ρ gives:

$$\frac{p}{\rho}+\frac{V^2}{2}+gz = \text{constant along the streamline}$$

Each term now has units of energy per unit mass:

$$\left[\frac{\text{J}}{\text{kg}}\right] = \left[\frac{\text{m}^2}{\text{s}^2}\right]$$

Where:

- p/ρ is the pressure related work per unit mass
- $V^2/2$ is the kinetic energy per unit mass
- gz is the gravitational potential energy per unit mass

Alternatively, we can divide through by γ to get:

$$\frac{p}{\gamma} + \frac{V^2}{2g} + z = \text{constant along the streamline}$$

Each term now has units of energy per unit weight: $LF/F = L$ (*head in units of length*)

Where:

- **Pressure head**: the height of a fluid column that would produce the pressure p at the base of the column
$$\frac{p}{\gamma} = h_p$$
- **Velocity head**: the height through which a fluid must fall to reach a final speed of V (neglecting frictional effects)
$$\frac{V^2}{2g} = h_v$$
- **Elevation head**: the height of the fluid particle relative to some reference point/datum
- **Total head**: the sum of the pressure, velocity, and elevation heads

$$h_{\text{total}} = h_p + h_v + z$$

Physical interpretation of the normal equation of motion

Unlike the Bernoulli equation along a streamline, the normal equation of motion describes the pressure and gravity needed to maintain a curved streamline.

$$p + \rho \int \frac{V^2}{\mathscr{R}}\, dn + \gamma z = \text{constant across the streamline}$$

Special case: Straight streamlines
If the streamlines are straight, then $\mathscr{R} \to \infty$, and thus the integral term goes to zero $V^2/\mathscr{R} \to 0$.

The normal equation of motion reduces to:

$$p + \gamma z = \text{constant across the streamline}$$

Or in between two points across streamlines:

$$p_1 + \gamma z_1 = p_2 + \gamma z_2$$

This is just the hydrostatic pressure distribution, even though the fluid is moving!

This happens because when there is no normal acceleration, the net force in the normal direction is zero, and thus the pressure gradient must balance the gravitational force.

3.5 Static, stagnation, dynamic, and total pressure

This section introduces different types of pressure associated with fluid motion, explains how kinetic energy can be converted into a pressure rise, and shows how Pitot-static tubes measure fluid velocity.

Types of pressure

Recall Bernoulli's equation along a streamline:

$$p + \frac{1}{2}\rho V^2 + \gamma z = \text{constant along the streamline}$$

We have the following pressures:

- **Static pressure**: the actual pressure of the fluid at a point in space

$$p$$

- **Dynamic pressure**: the pressure representing the kinetic energy per unit volume of the fluid. It is also equivalent to the pressure rise that would occur if the fluid were brought to rest without frictional losses (isentropically).

$$\frac{1}{2}\rho V^2$$

- **Stagnation pressure**: the pressure at a point in the fluid where the velocity is zero (stagnation point). It is the sum of the static and dynamic pressures.

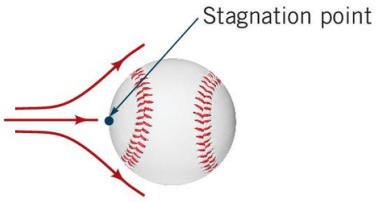
$$p_0 = p + \frac{1}{2}\rho V^2$$

- **Total pressure**: the sum, or the total ... pressure... (yeah)

$$p + \frac{1}{2}\rho V^2 + \gamma z$$

All four pressures have units of ... pressure ... (force per unit area).

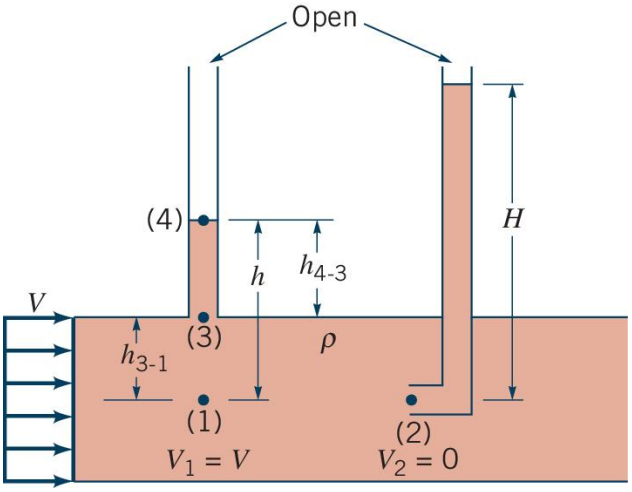
Note that for any stationary body placed into a flowing liquid, there would be a stagnation point that divides where the flow splits into two paths around the body.



If elevation effects are negligible ($\gamma z \approx 0$), then the stagnation pressure would be the largest possible pressure along a given streamline. This is because it represents the conversion of all the kinetic energy into a pressure rise.

Note on Static Pressure Measurement

Consider the following set up where we measure the static pressure within a flowing fluid using a **piezometer tube**:



To find the static pressure at point 1, we can use the following manometer relations:

- From 3.4, we know that for straight streamlines, the pressure distribution is hydrostatic, thus

$$p_1 = p_3 + \gamma h_{3-1}$$

- Using manometer rules, we know that the pressure at point 3 is given by:

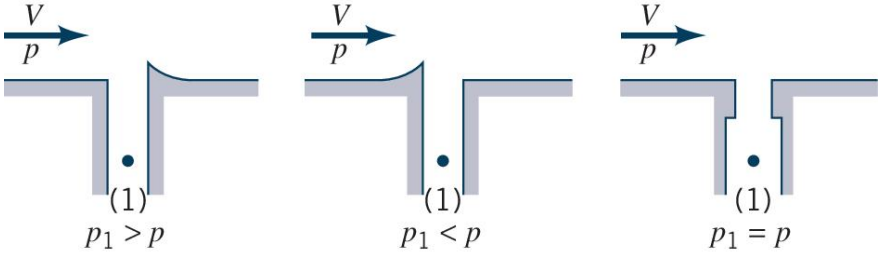
$$p_3 = \gamma h_{4-3}, \quad \text{(gage pressure)}$$

- Thus, putting it together:

$$p_1 = \gamma h_{4-3} + \gamma h_{3-1} = \gamma h$$

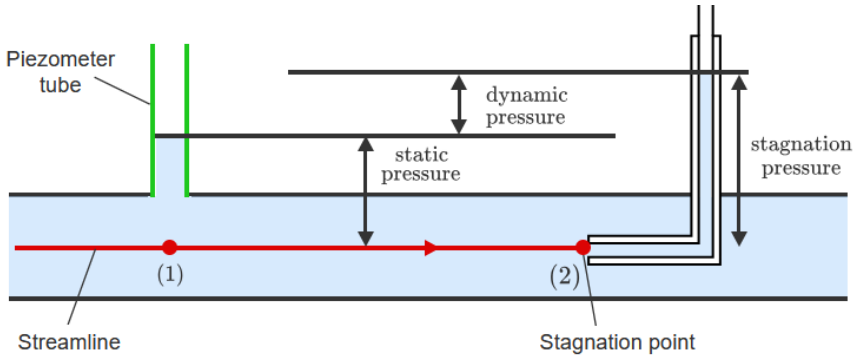
  Where h is the height from point 1 to the free surface of the piezometer tube.

Note: The piezometer tube must have a smooth hole with no burrs or imperfections. Otherwise, some of the fluid's kinetic energy will be converted into a pressure rise at the point of measurement.



Static vs Dynamic Pressure

The static and dynamic pressure can be visualized below:



$$p_1 + \frac{1}{2}\rho V_1^2 + \gamma z_1 = p_2 + \frac{1}{2}\rho V_2^2 + \gamma z_2$$

$$p_1 + \frac{1}{2}\rho V_1^2 = p_2$$

Static pressure + Dynamic Pressure = Stagnation Pressure

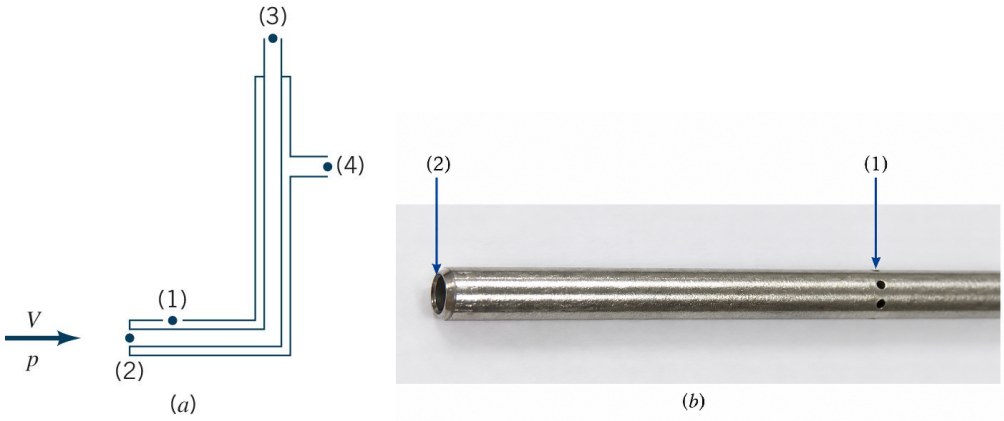
Note that the height difference (using notation from the previous example) between the static pressure and stagnation pressure is given by:

$$H - h = \frac{V^2}{2g}$$

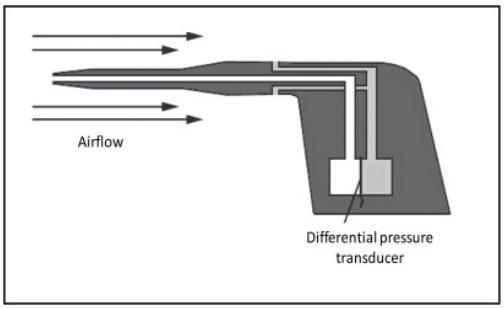
This is the velocity head from section 3.4 !

Pitot-static tube

By knowing both the static and stagnation pressures, we can calculate the velocity of the fluid at a point. It is composed of two concentric tubes, whose pressures p_3 and p_4 are measured separately or as a difference $p_3 - p_4$ with a differential gage.



Unrelated to the problem set up shown previously, this shows the two concentric tubes of a Pitot-static tube a bit better:



Back to the problem. Assuming negligible elevation changes between points 1,2,3,4:

The center tube measures the stagnation pressure at point 2.

$$p_3 = p + \frac{1}{2}\rho V^2$$

Where p and V are the static pressure and velocity of the fluid upstream from point 2.

The outer tube is made of several holes at an appropriate distance from the tip (so that the streamlines are straight) to measure the static pressure at point 1.

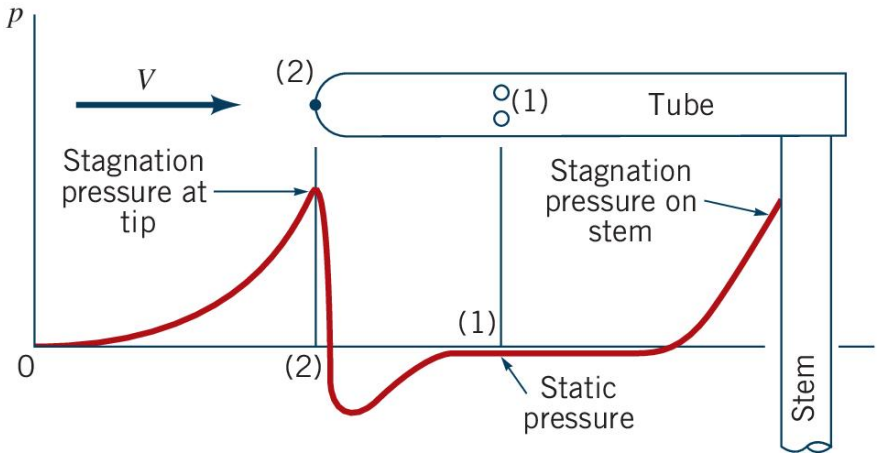
$$p_4 = p_1 = p$$

Assuming negligible elevation changes and properly positioned static pressure holes, the measured static pressure equals the undisturbed upstream static pressure.

Combining the two equations gives:

$$p_3 - p_4 = \frac{1}{2}\rho V^2$$

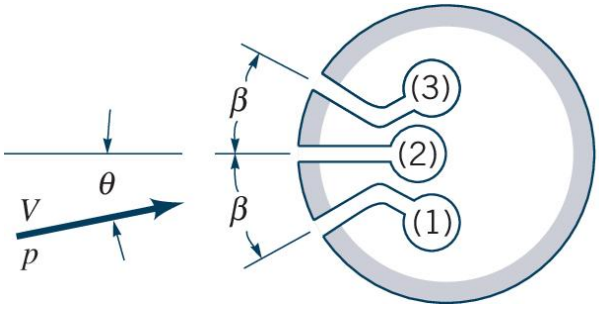
$$\boxed{V = \sqrt{\frac{2(p_3 - p_4)}{\rho}}}$$



Directional Pitot Tubes

Since we are finding the stagnation pressure at a point along the streamline, the Pitot tube must be aligned with the flow. Typically, yaw angles up to $12 - 20°$ depending on the design of the Pitot tube are acceptable (¡1% error). However, for larger yaw angles, the Pitot tube must be aligned with the flow to get an accurate measurement of the stagnation pressure.

Consider the following example of a directional Pitot tube:



This can be rotated to align with the flow until the side hole pressures are equal: $p_1 = p_3$. When the center hole faces the flow, the flow and pressure distribution are symmetric about the centerline (stagnation point).

Having found the angle such that $\theta = 0$, we can then calculate the velocity of the flow like before.

3.6 Examples of use of the Bernoulli equation

This section applies Bernoulli's equation to three major types of flow: free jets, confined flows, and flow rate measurement. It also introduces the vena contracta effect, cavitation, and the continuity equation.

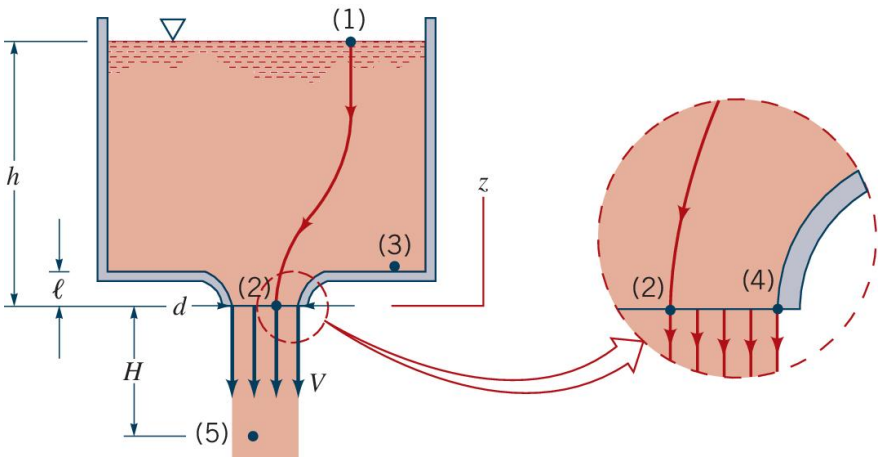
The general approach for a steady, inviscid, incompressible flow along a streamline is to apply Bernoulli's equation between two points along the streamline:

$$p_1 + \frac{1}{2}\rho V_1^2 + \gamma z_1 = p_2 + \frac{1}{2}\rho V_2^2 + \gamma z_2$$

Typically, we need to use additional equations to solve for the unknowns, such as the continuity equation (conservation of mass). More importantly, we need to be able to identify two convenient points along a streamline and identifying which terms can be simplified or neglected.

Free Jets

A free jet is fluid discharged into a surrounding medium, typically the atmosphere, without being confined by solid walls.



Applying Bernoulli's equation between points 1 (free surface) and 2 (nozzle exit), we have the following simplifications:

- Both points are exposed to the atmosphere, so $p_1 = p_2 = 0$ (gage pressure)
- The reservoir area is much larger than the nozzle area, so the velocity at the free surface is negligible $V_1 \approx 0$
- Define the nozzle exit as the elevation datum, so $z_1 = h$ and $z_2 = 0$

Thus, Bernoulli's equation reduces to:

$$\gamma h = \frac{1}{2}\rho V_2^2 \implies \boxed{V_2 = \sqrt{2gh}}$$

This is also known as Torricelli's law. This is also the same result found by converting the potential energy of the fluid in the reservoir into kinetic energy at the nozzle exit.

Note: The velocity at the nozzle exit can also be found using points 2 and 3. Using the following:

- $z_3 = \ell$
- $V_3 = 0$ since it is far from the nozzle exit. Near the nozzle, streamlines converge and the fluid accelerates. Farther away, the flow is much slower.
- $p_3 = \gamma(h - \ell)$ from hydrostatic pressure distribution

Why is exit pressure atmospheric?
Consider the normal equation of motion from section 3.3:

$$-\gamma\frac{dz}{dn} - \frac{\partial p}{\partial n} = \rho\frac{V^2}{\mathscr{R}}$$

or its integral form:

$$p + \rho\int\frac{V^2}{\mathscr{R}}\,dn + \gamma z = \text{constant across the streamline}$$

Since the streamlines are straight at the nozzle exit, $\mathscr{R} \to \infty$, and thus the integral term goes to zero. Also, the height z is constant across the nozzle exit, so $dz/dn = 0$ and $\gamma z =$ constant. Thus, the normal equation of motion reduces to:

$$\frac{\partial p}{\partial n} = 0$$

$$p = \text{constant across the nozzle exit}$$

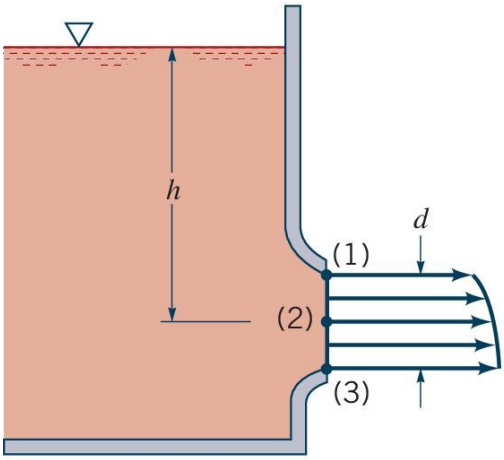
We know that the pressure at the surface of the jet is atmospheric, and since pressure is *constant across the nozzle exit*, the pressure at the center of the jet must also be atmospheric. Thus, $p_2 = 0$ (gage pressure).

Note: For straight horizontal streamlines, z is constant along each streamline, but varies across streamlines, so $dz/dn \neq 0$. Thus, the pressure across the jet follows the hydrostatic pressure distribution $p + \gamma z = C$. However, since the top and bottom of the jet are exposed to the atmosphere and the nozzle diameter is small (small changes in z), the pressure at the center of the jet is approximately equal to atmospheric pressure.

We can see this in how the streamlines of a horizontal free jet will tend to bend downwards due to gravity, thus resulting in a finite $\mathscr{R}$ even if the pressure is constant (zero pressure gradient $\partial p/\partial n = 0$).

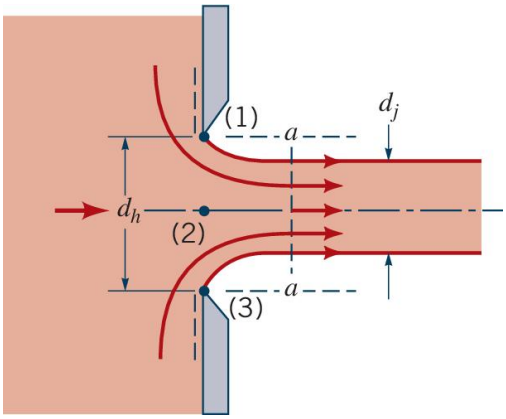
Ven Contracta Effect

Continuing our lovely discussion on horizontal free jets, consider the following example:



The fluid velocity across the jet is not uniform with $V_1 < V_2 < V_3$, where V_2 is the velocity at the center of the jet. This is due to elevation changes across the jet. In general, since $d << h$, we can call V_2 the approximate average velocity of the jet.

If the exit is not a smooth, well-contoured nozzle, but rather a sharp-edged opening, the diameter of the jet d_j will be smaller than the diameter of the hole d_h. This is known as the **vena contracta effect**. This is a result of the inability of the fluid to make sharp right turns, and instead needing to follow a curved path.



Since the streamlines at the exit hole are curved (finite $\mathscr{R}$), the pressure across them is not constant. Recall the normal equation of motion:

$$-\gamma\frac{dz}{dn} - \frac{\partial p}{\partial n} = \rho\frac{V^2}{\mathscr{R}}$$

For negligible graviatational effects, $\gamma \approx 0$, and thus the pressure gradient is given by:

$$\frac{\partial p}{\partial n} = -\rho\frac{V^2}{\mathscr{R}}$$

It can be seen that it would take an infinite pressure gradient across the streamlines to make the fluid turn a sharp right angle.

$$\mathscr{R} \to 0 \implies a_n = \frac{V^2}{\mathscr{R}} \to \infty \implies \left|\frac{\partial p}{\partial n}\right| \to \infty$$

Thus, a finite pressure gradient cannot produce the infinite acceleration needed for an instantaneous sharp turn. The streamlines must curve.

Why is pressure highest at the centerline?
At the exit, there is a finite radius of curvature $\mathscr{R}$, with corresponding centers of curvature outside the jet. Similar to the argument for horizontal flow in section 3.3, the pressure gradient is negative, and thus for increasing n (towards the center of curvature, away from the centerline), the pressure decreases. Thus, the pressure is highest at the centerline of the jet (p_2). The lowest pressure is at the edge of the jet, where the pressure is equal to the atmospheric pressure ($p_1 = p_3 = 0$). This is consistent with our argument for *straight horizontal streamlines*, where the pressure distribution is hydrostatic, and the pressure is highest at the centerline.

What happens at the vena contracta?
Further downstream, after section $a - a$, the streamlines straighten out and reaches its minimum diameter d_j. The radius of curvature $\mathscr{R} \to \infty$, and thus the pressure gradient across the streamlines goes to zero, leaving only the gravitational effects $\gamma dz/dn$. Neglecting gravitational effects, the pressure across the streamlines is constant.

Contraction Coefficient

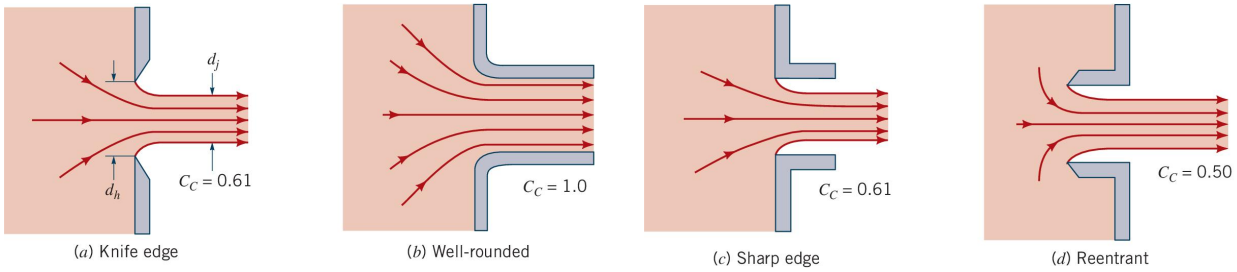
Okay, enough with the theory. Let's quantify the vena contracta effect. The contraction coefficient C_c is defined as the ratio of the cross-sectional area of the jet at the vena contracta A_j to the hole A_h:

$$\boxed{C_c = \frac{A_j}{A_h}}$$

For circular cross-sections, this can be expressed in terms of diameters:

$$\boxed{C_c = \frac{d_j^2}{d_h^2}}$$

Experimentally, the coefficients were found for some common geometries:



(a) Knife edge (b) Well-rounded (c) Sharp edge (d) Reentrant

Confined Flows

Unlike free jets, confined flows are constrained by solid walls, such as in pipes and nozzles. For these situations it is necessary to use the conservation of mass (the continuity equation) along with the Bernoulli equation.

Conservation of mass: Continuity equation

The continuity equation states that the mass flow rate into and out of a control volume must be equal for steady flow (no accumulation of mass, everything that goes in must come out). The mass flowrate is given by:

$$\dot{m} = \rho \cancel{\dot{V}} = \rho A V$$

Where $[\dot{m}] = \frac{\text{slug}}{\text{s}}, \frac{\text{lbm}}{\text{s}}, \frac{\text{kg}}{\text{s}}$ and $[\cancel{\dot{V}}] = \frac{\text{ft}^3}{\text{s}}, \frac{\text{m}^3}{\text{s}}$ are the mass and volumetric flowrates, respectively.

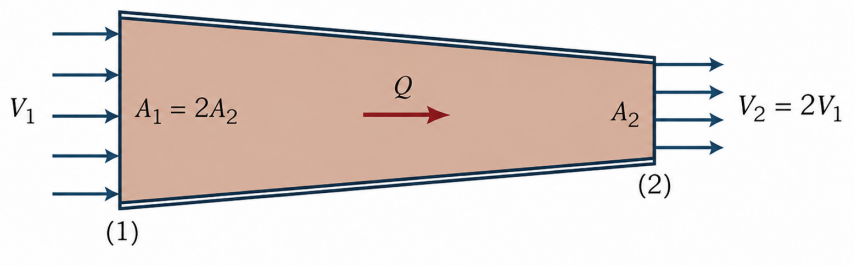
A is the cross-sectional area of the flow, and V is the average velocity of the flow normal to the cross-section.

For steady flow with one inlet and one outlet:

$$\boxed{\rho_1 A_1 V_1 = \rho_2 A_2 V_2}$$

For incompressible flow, $\rho_1 = \rho_2$, and thus we can write:

$$\boxed{A_1 V_1 = A_2 V_2}$$



Physically, this means that if the cross-sectional area of the flow decreases, the average velocity must increase to maintain a constant mass flowrate. Recalling the Bernoulli equation along a streamline for horizontal flow ($z_1 = z_2$):

$$p_1 + \frac{1}{2}\rho V_1^2 = p_2 + \frac{1}{2}\rho V_2^2$$

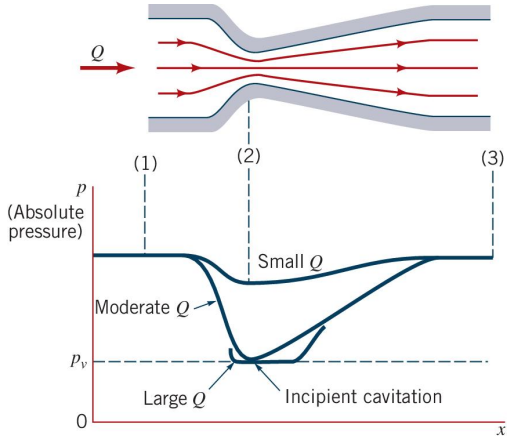
If the velocity increases, the pressure must decrease! So smaller flow area produces higher velocity and lower static pressure (total energy is conserved or smth).

Cavitation

Since an increase in velocity results in a decrease in pressure, if the differences in velocity are considerable, then the differences in pressure can be considerable as well (big). For gasses, this could lead to compressibility effects, but for liquids, this could lead to cavitation.

Cavitation occurs when the liquid pressure drops below the vapor pressure and the liquid "boils".

When the fluid then moves into a higher pressure region (lower velocity), the vapor bubbles can collapse violently, causing damage to the surrounding solid surfaces.

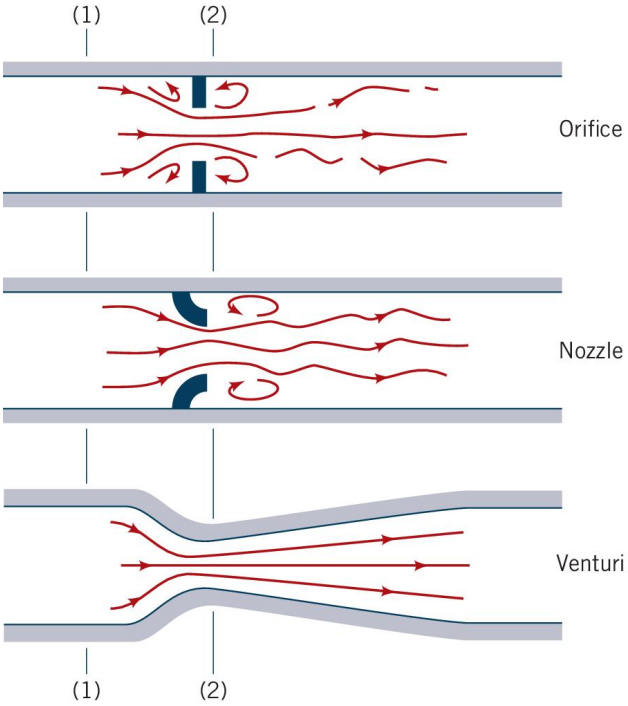


IMPORTANT: Use absolute pressure when comparing a liquid's pressure with its vapor pressure!

Flowrate measurement

Flowmeters use the relationship between velocity and pressure to measure the flowrate through a pipe. One such device is the pitot-static tube from section 3.5.

A different approach to the pitot-static tube is to instead place some type of restriction within the pipe and to measure the pressure difference between the low-velocity, high-pressure upstream section (1) and the high-velocity, low-pressure downstream section (2). Below are some common types of flowmeters that use this approach:



Flowmeter Equation

For horizontal flow, we can apply Bernoulli's equation between points 1 and 2:

$$p_1 + \frac{1}{2}\rho V_1^2 = p_2 + \frac{1}{2}\rho V_2^2$$

Rearranging gives:

$$p_1 - p_2 = \frac{1}{2}\rho(V_2^2 - V_1^2)$$

From the continuity equation:

$$V_1 = \frac{A_2}{A_1}V_2$$

Plug in:

$$p_1 - p_2 = \frac{1}{2}\rho V_2^2\left(1 - \frac{A_2^2}{A_1^2}\right)$$

Solve for V_2 and use $\dot{\forall} = Q = A_2 V_2$ to get the flowrate:

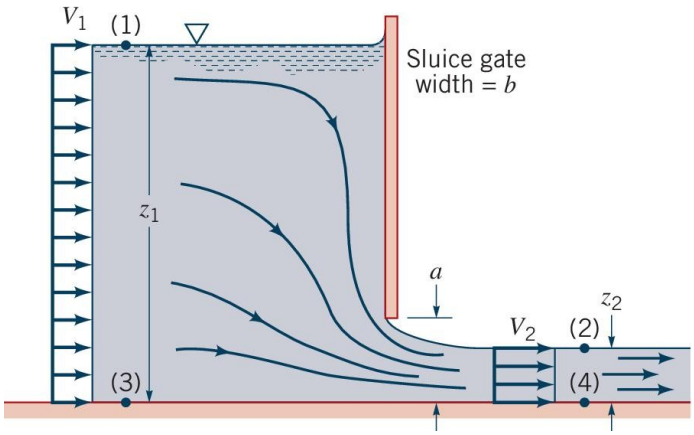
$$\boxed{Q = A_2 V_2 = A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho\left(1 - \left(\frac{A_2}{A_1}\right)^2\right)}}}$$

Thus, the ideal flowmeter for a fixed geometry and fluid density has the relationship: $Q \propto \sqrt{\Delta p}$.

Flowrate measurement in open channels

We can also derive the flowrate relationship for open channel flow (not quite free jet or confined flow). Two of these devices, the sluice gate and the sharp-crested weir, are discussed next under the assumption of steady, inviscid, incompressible flow.

Sluice Gate



Applying Bernoulli's equation between free surfaces at 1 and 2:

$$p_1 + \frac{1}{2}\rho V_1^2 + \gamma z_1 = p_2 + \frac{1}{2}\rho V_2^2 + \gamma z_2$$

If the gate and channel width are equal, then $A_1 = bz_1$ and $A_2 = bz_2$, and thus from the continuity equation:

$$Q = A_1 V_1 = A_2 V_2 \implies bV_1 z_1 = bV_2 z_2$$

With $p_1 = p_2 = 0$ (gage pressure), we get:

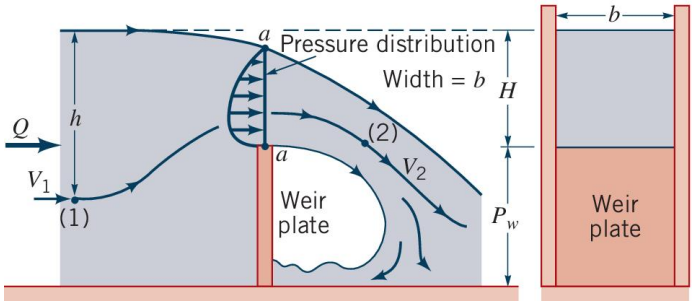
$$\boxed{Q = z_2 b \sqrt{\frac{2g(z_1 - z_2)}{1 - \left(\frac{z_2}{z_1}\right)^2}}}$$

If $z_1 >> z_2$, then $z_2/z_1 \approx 0$, and thus we can simplify to:

$$\boxed{Q \approx z_2 b\sqrt{2g(z_1 - z_2)}}$$

Note: The downstream depth z_2, not the gate opening a was used because of the vena contracta effect. $C_c = z_2/a \approx 0.61$ for a sluice gate, this is valid for the range $0 < a/z_1 < 0.2$.

Weird Weir



Its weird, so we cannot calculate the flowrate accurately, what can ya do?

Instead, we can assume that weir flow is similar to free jet flow, where the *average* velocity is proportional to $V_{avg} \propto \sqrt{2gH}$ and the flow area for this rectangular weir to be $A = bH$. Thus, we can write:

$$Q = C_1 H b \sqrt{2gH} = C_1 b \sqrt{2g}\, H^{3/2}$$

Where C_1 is an unknown constant. (So fucking weird)

3.7 The energy line and the hydraulic grade line

This section introduces two graphical interpretations of Bernoulli's equation: the energy line (EL) and the hydraulic grade line (HGL).

Definitions

- **Energy line (EL)**: a line that represents the total energy of the fluid at a point in space, the same as the total head in Bernoulli's equation.

$$EL = H = \frac{p}{\gamma} + \frac{V^2}{2g} + z$$

- **Hydraullic grade line (HGL)**: a line that represents the pressure and elevation heads of the fluid at a point in space. It is the energy line minus the velocity head.

$$HGL = \frac{p}{\gamma} + z$$

$$EL - HGL = \frac{V^2}{2g}$$

Energy Line

For steady, inviscid, incompressible flow along a streamline, the energy line is constant ($H = \text{constant}$).

However, viscous losses (friction) and other energy losses can cause the energy line to drop along the streamline.

Hydraullic Grade Line

Rewriting for hydraulic grade line in terms of the energy line:

$$HGL = EL - \frac{V^2}{2g}$$

Since the energy line is constant for an ideal flow, the hydraulic grade line is dependent on the velocity head. Thus:

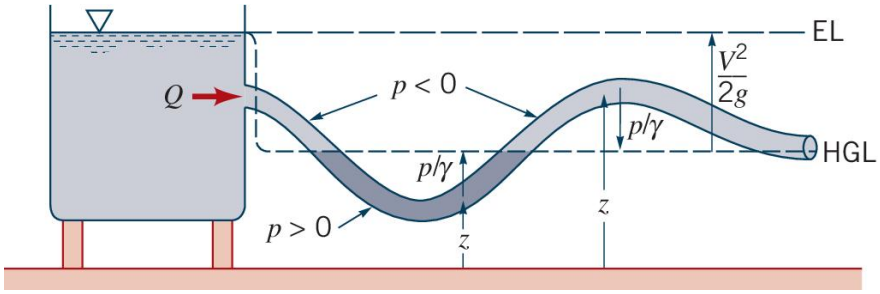
- Increasing velocity lowers the HGL
- Decreasing velocity raises the HGL
- Constant velocity keeps the HGL a constant offset below the EL, assuming ideal flow

Rewriting for pressure along a streamline in terms of the hydraulic grade line:

$$\frac{p}{\gamma} = HGL - z$$

Thus, the vertical distance between the streamline elevation and the hydraulic grade line is equal to the pressure head. This means that:

- If streamline is below the HGL, then $p > 0$
- If streamline is on the HGL, then $p = 0$
- If streamline is above the HGL, then $p < 0$

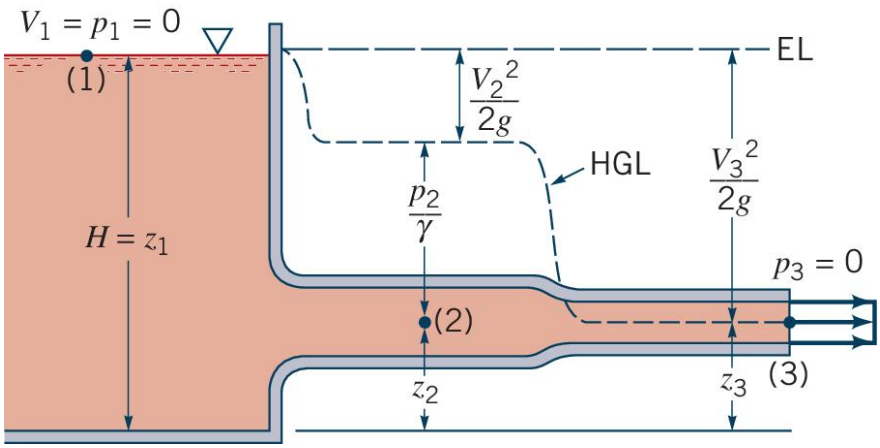


How the lines are measured

- Pitot-static tube: measures stagnation pressure vs elevation, allowing the energy line elevation to be determined

- Piezometer tube: measures static pressure head, allowing the hydraulic grade line elevation to be determined

Example:



Note: Review Example 3.7.1

3.8 Restrictions on use of the Bernoulli equation

We are super familiar with the Bernoulli equation for steady, inviscid, incompressible flow along a streamline:

$$p + \frac{1}{2}\rho V^2 + \gamma z = C$$

Compressibility

While assuming incompressible flow is valid for most liquids, it is generally not valid for gasses where density can change significantly with pressure.

From the definition of the bulk modulus E_ν:

$$\frac{d\rho}{\rho} = \frac{dp}{E_\nu}$$

For a small local change in density, we can write:

$$\frac{\Delta\rho}{\rho} = \frac{\Delta p}{E_\nu}$$

If we define the (arbitrary) criteria that an 4.5% density change is significant, we can model the fluid as incompressible when:

$$\frac{\Delta\rho}{\rho} \leq 0.045$$

Isothermal Flow

When the temperature of an ideal gas is constant along a streamline, $E_\nu = p$, and thus the

incompressible flow assumption is valid when:

$$\frac{\Delta\rho}{\rho} = \frac{\Delta p}{p} \leq 0.045$$

This criterion should be used with caution because most isothermal flows are accompanied by viscous effects.

Isentropic Flow
When an ideal gas has constant entropy along a streamline (no heat transfer or friction), $E_\nu = kp$, where k is the ratio of specific heats. Thus, the incompressible flow assumption is valid when:

$$\frac{\Delta\rho}{\rho} = \frac{\Delta p}{kp} \leq 0.045$$

For air, $k = 1.4$:

$$\implies \frac{\Delta p}{p} \leq (0.045)(1.4) = 0.063$$

Note that the Bernoulli equation derived for steady, inviscid flow is an isentropic process! Thus, we can relate the difference in pressure (with negligible elevation change) to the fluid velocity:

$$\Delta p = \frac{1}{2}\rho V^2$$

$$\implies \frac{\rho V^2}{p} \leq 0.126$$

For air at standard conditions, this corresponds to a velocity of $V \leq 102\frac{\text{m}}{\text{s}} \approx 350\frac{\text{ft}}{\text{s}}$ or $\text{Ma} \leq 0.3$. Thus, the incompressible flow assumption is valid for air at standard conditions when it is moving sufficiently slow compared to the speed of sound.

Note: USE ABSOLUTE PRESSURES, THESE ARE IN TERMS OF IDEAL GASES

Unsteady Effects

Recall the full streamwise acceleration is given as:

$$a_s = \frac{\partial V}{\partial t} + V\frac{\partial V}{\partial s}$$

For steady flow,

$$\frac{\partial V}{\partial t} = 0$$

However, for unsteady flow, we will need to account for the local acceleration term $\partial V/\partial t$. Thus, if we repeat the derivation for Newton's second law along a streamline keeping the local acceleration term, we will get a modified Bernoulli equation:

$$\rho\frac{\partial V}{\partial t}ds + dp + \frac{1}{2}\rho\, d(V^2) + \gamma dz = 0 \quad \text{(along a streamline)}$$

For an incompressible flow between two points along a streamline, we can integrate to get:

$$p_1 + \frac{1}{2}\rho V_1^2 + \gamma z_1 = p_2 + \frac{1}{2}\rho V_2^2 + \gamma z_2 + \rho \int_{s_1}^{s_2} \frac{\partial V}{\partial t}\, ds \quad \text{(along a streamline)}$$

Note:

- If flow conditions change slowly with time, then the local acceleration term is negligible, and steady flow is a good approximation.
- Reference frame: a flow that is unsteady in one reference frame may be steady in another reference frame. Look at example 3.8.2

Rotational Effects

Care must be used in applying the Bernoulli equation across streamlines.

For steady, inviscid, incompressible flow, Bernoulli applies along an individual streamline (our old friend):

$$p + \frac{1}{2}\rho V^2 + \gamma z = C$$

However, different streamlines may have different constants

$$C_1 \neq C_2$$

(so we cannot apply Bernoulli across streamlines without additional considerations).

If the flow is also irrotational (the fluid particles do not "spin" as they move), the Bernoulli constant is the same throughout the connected flow region:

$$C_1 = C_2 = C$$

However, if the flow is rotational (the fluid particles spin as they move), then the Bernoulli equation is restricted to flow along a streamline.

Mathematically, the flow is irrotational if the curl of the velocity field is zero:

$$\nabla \times \vec{V} = 0$$

Other Restrictions

Viscous Effects

Mentioned briefly previously for energy line in section 3.7, when viscous effects are significant, mechanical energy is dissipated and the total head is not constant along a streamline.

$$H_1 \neq H_2$$

Pumps and Turbines

Like the issue with viscous effects, pumps and turbines add or remove energy from the flow,

and thus the total head is not constant along a streamline.

Pumps add energy to the flow, and thus the total head increases along a streamline. Turbines extract energy from the flow, and thus the total head decreases along a streamline.